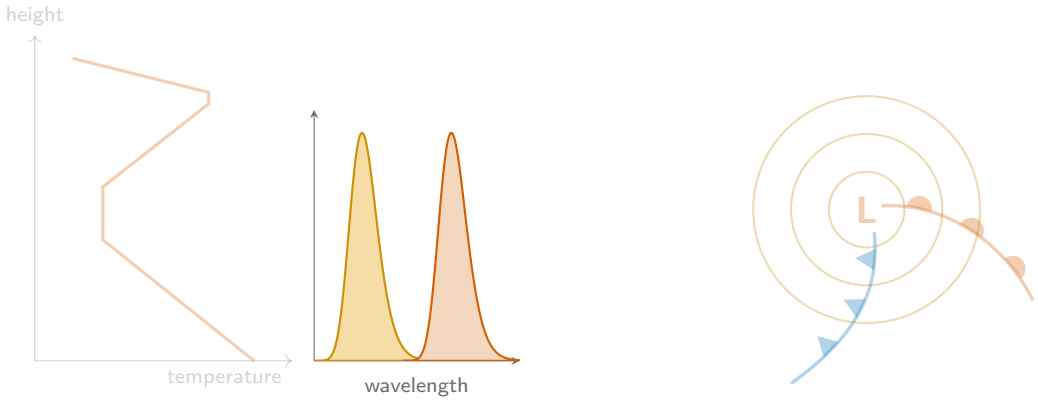
The Atmosphere: Part II

Solar radiation: radiation and heating

A worked, step-by-step guide with figures, examples and answers



Part I Part II Part III

Part II of The Atmosphere

One part of The Atmosphere, with the same numbering as the full book. Every question, data table, figure and worked answer is on the page.

Ahmad Ammar ▪ School of Geography and Environmental Sciences, Ulster University

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Chapter 1

How to use this book

1.1 What is in it

This book teaches the atmosphere through questions you work yourself. It has three parts, each following on from the one before: Part I describes the structure of the atmosphere, Part II explains how the Sun heats it, and Part III how that uneven heating fills it with moisture and sets it in motion. Everything you need is on the page: every question, data table, map and figure, with its worked answer.

Table 1.1. The three parts of the book.

Part	What it covers	Chapters	Appendix
I	**Vertical structure of the atmosphere.** Pressure and the halving rule, oxygen at altitude, the standard atmosphere, soundings, inversions and the tropopause, lapse rates, stability, climates.	2–10	A
II	**Solar radiation.** The laws of radiation, black bodies and planetary temperatures, the Earth's energy balance and the greenhouse effect, sun angle and the seasons.	11–19	B
III	**Moisture, motion and weather maps.** Humidity and the dew point, the forces that move the air, the geostrophic wind and friction, station models, streamlines, fronts and a full weather-map analysis.	20–30	C

Every part opens with a chapter that maps each question to the chapter that answers it, and closes with a chapter that gathers all the answers in one table, a list of mistakes to avoid and a self-test. Its appendix holds the formulas, derivations and symbols. Appendix D lists the shared constants and units, and the references are at the end.

1.2 How each chapter is laid out

1. **The ideas first,** in plain language. Every important equation is numbered, for example (3.4), and followed by a *where* list of its symbols, units and values.
2. **The question**, in a box, with its data beside it.
3. **The method**, one step per line, with every number and unit shown.
4. **The answer**, in a green box, followed by a **check** that it is sensible.
5. **Worked examples**, **key ideas**, points to **take care** over, and **questions that come up**.
6. **Try it yourself**: a practice set, with answers in the green box that follows.
7. **Python**: the equations as functions, which you can run and change. They are also collected in one tested package.

How to get the most out of the book

Cover the green answer box, work the question on paper, then uncover it. If your answer differs, find the step where you diverged before reading on. That step is where the learning is.

1.3 Conventions used throughout

- **Numbering.** "Eq. (4.2)" is equation 2 of Chapter 4; figures and tables are numbered the same way. "§ 4.3" is section 3 of Chapter 4. Appendices are lettered A to D, one per part and one for shared constants.
- **Units** are SI, written as powers: $\mathrm{W\,m^{-2}}$ means watts per square metre, $\mathrm{m\,s^{-1}}$ metres per second. Pressure is in hectopascals, which are the same as millibars ($1\ \mathrm{hPa} = 1\ \mathrm{mb} = 100\ \mathrm{Pa}$). Heights are in metres or kilometres.
- **Temperature.** Temperatures in the data are in degrees Celsius. Any formula in which temperature is multiplied, divided or raised to a power (the gas law, the radiation laws, the potential temperature) needs kelvin: $T_\mathrm{K} = T_{\circ\mathrm{C}} + 273.15$. A *difference* of temperature is the same in both scales, so lapse rates can be written in $^\circ\mathrm{C\,km^{-1}}$ or $\mathrm{K\,km^{-1}}$.
- **Constants.** Each part uses rounded values, stated where they first appear; Appendix D gives precise values and sources.
- **Rounding.** Intermediate results keep at least four significant figures; answers are rounded at the end, usually to three or four.
- **Cross-references between parts** are written in the same way as within a part, with the part named: "Part II, § 17.7".
- **References** to the literature are given as author and year, for example Stull (2017), and listed in full at the end.

1.4 What to have beside you

A calculator with powers, logarithms and trigonometric functions, set to **degrees**; graph paper, a ruler, a soft pencil and three or four coloured pencils, so that each line on a graph has its own colour. Python is optional.

1.5 Acknowledgement

The questions and data in this book grew out of the practicals of EGM305 The Atmosphere at Ulster University, and are reused with the permission of the module staff. The explanations, figures, code and any errors are the author's.

1.6 Reading

Good textbooks for going further are Lutgens et al. (2019) and Ahrens and Henson (2022). A free textbook with worked examples is Stull (2017). For more depth, Wallace and Hobbs (2006) treats the physics at the next level up.

Part II

Solar radiation

Chapter 11

Part II at a glance

11.1 What this part covers

This part has three goals: the *laws of radiation*, *black bodies and radiant energy*, and the *determination of solar intensity*. The five questions build on each other, so it pays to work through them in order. They also build on Part I: the same atmosphere whose layers you plotted there (Chapter 5) is now heated by the Sun and cooled by its own radiation.

The laws of radiation. Every object gives off radiation, and its temperature decides both *what kind* (the wavelength of its peak, Wien's law, Question 1) and *how much* (the total, the Stefan–Boltzmann law, Question 2).

Black bodies and planets. You plot the spectra of three planets, read their temperatures from the peaks (Questions 3a and 3b), calculate the temperature each planet *should* have from its distance to the Sun and its reflectivity (Question 3c), match the spectra to the planets (Question 3d) and explain why the real temperatures differ (Question 3e).

The Earth's energy balance and solar intensity. You calculate the temperature the Earth would have without an atmosphere, which measures the greenhouse effect (Question 4), and then show why the angle of the Sun makes the tropics warm, the poles cold and the seasons change (Question 5).

Table 11.1. Where each question is answered, which equations it needs, and where to read more (Lutgens et al., 2019; Ahrens and Henson, 2022).

Question	What it asks	Chapter	Equations	Texts
1	the Sun's temperature from its peak wavelength	13	(13.1)	ch. 2
2a–2e	emission of the Sun and the Earth; why T and E ratios differ	14	(14.1), (14.2)	ch. 2
3 (graph), 3a–3b	plot three spectra; peaks and temperatures	15	(13.2)	ch. 2
3c–3e	planetary temperatures and why they differ	16	(16.2), (16.1)	ch. 2
4a–4g	albedo, energy balance, greenhouse effect	17	(17.1)–(17.3)	ch. 2
5a–5c	sun angle, beam spreading, seasons	18	(18.2)–(18.3)	ch. 2–3

Chapter 12 builds the physics that every question uses; Chapter 19 gathers all the answers in one table. The formulas, derivations, recomputed data, symbols and constants for this part are in Appendix B.

11.2 Conventions for this part

The conventions of the whole book are in § 1.3. Three more apply here.

- **Temperatures in the radiation laws are always in kelvin.** The worked answers in this part convert with $T_{\circ\mathrm{C}} = T_{\mathrm{K}} - 273.16$; the exact offset is 273.15 (Tiesinga et al., 2021). The difference of 0.01 K never changes an answer at the precision used here.

- **Constants.** Every answer uses the rounded values Wien's constant $C = 2898\,\mu\mathrm{m\,K}$ and the Stefan–Boltzmann constant $\sigma = 5.67 \times 10^{-8}\,\mathrm{W\,m^{-2}\ K^{-4}}$. Precise values and sources are listed in Table B.3.
- **Units** are written as powers: $\mathrm{W\,m^{-2}}$ means watts per square metre. Wavelengths are in micrometres, $1\,\mu\mathrm{m} = 10^{-6}\,\mathrm{m}$.

Chapter 12

What radiation is

This chapter builds the ideas that every question of this part uses. It answers no question directly; Chapters 13 to 18 refer back to it.

12.1 Energy that needs no material to travel

Energy moves from place to place in three ways. *Conduction* passes it from molecule to molecule through contact (a spoon warming in hot tea). *Convection* carries it in a moving fluid (warm air rising from a radiator). *Radiation* needs no material at all: it is energy travelling as an electromagnetic wave, a ripple of electric and magnetic fields (Lutgens et al., 2019, ch. 2). Between the Sun and the Earth there are 150 million kilometres of almost empty space, so radiation is the only way the Sun's energy can reach us.

Two properties of radiation matter here:

- **It carries energy.** The body that emits it loses energy; the body that absorbs it gains energy and warms.
- **It comes in many wavelengths.** Radio waves, microwaves, infrared, visible light, ultraviolet and X-rays are all the same kind of thing. They differ only in wavelength.

A question that comes up

Is this the same as radioactivity?
No. In meteorology "radiation" always means electromagnetic radiation: sunlight, the infrared glow of the ground, and so on. Radioactivity is the decay of unstable atomic nuclei, an entirely different process. Only the word is shared.

12.2 Wavelength, frequency and speed

A wave is a repeating pattern of crests and troughs. The **wavelength** λ (the Greek letter lambda) is the distance from one crest to the next (Figure 12.1). The **frequency** f is the number of crests that pass a point each second. Shake one end of a rope slowly and the crests are far apart; shake it quickly and they bunch up.

All electromagnetic waves travel through space at the same speed, the speed of light c. In one second a source sends out f crests, each λ apart, and together they cover the distance the wave travels in one second:

$$c = f\lambda. \tag{12.1}$$

where

c	speed of light ($\mathrm{m\,s^{-1}}$)	3.00×10^8, Table B.3
f	frequency: crests per second ($\mathrm{s^{-1}}$, or hertz, Hz)	calculated
λ	wavelength (m)	given or calculated

Because c is fixed, frequency and wavelength are tied together: a short wavelength means a high frequency, and a long wavelength a low one.

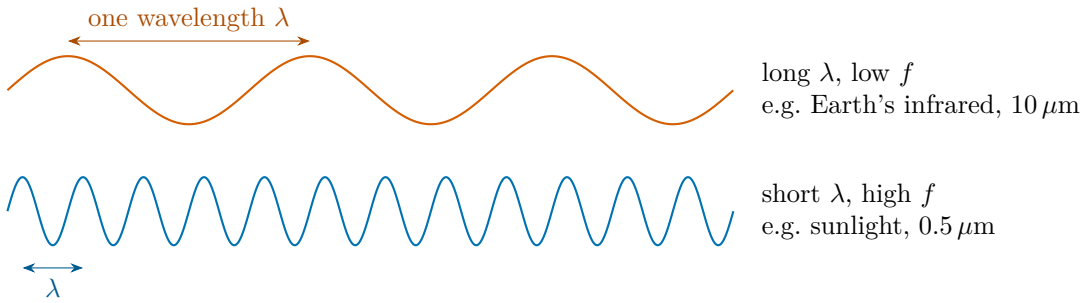


Figure 12.1. Wavelength and frequency. Both waves move at the speed of light, so the one whose source oscillates faster (blue) has its crests closer together, Eq. (12.1). Not to scale: the crests of the Earth's infrared are about 10 μm apart, a seventh of the width of a hair.

Worked example: the frequencies of sunlight and of the Earth's glow

Rearranging Eq. (12.1), $f = c/\lambda$. For the Sun's peak wavelength of $0.5\,\mu\text{m} = 0.5 \times 10^{-6}$ m:

$$f_{\text{Sun}} = \frac{3.00 \times 10^{8}}{0.5 \times 10^{-6}} = 6.0 \times 10^{14}\ \text{Hz}.$$

For the Earth's peak near $10\,\mu\text{m} = 10 \times 10^{-6}$ m:

$$f_{\text{Earth}} = \frac{3.00 \times 10^{8}}{10 \times 10^{-6}} = 3.0 \times 10^{13}\ \text{Hz}.$$

Sunlight oscillates twenty times faster than the Earth's infrared, because its wavelength is twenty times shorter.

12.3 Photons: radiation comes in packets

Radiation is emitted and absorbed in separate packets of energy called **photons**, much as rain arrives as separate drops. The energy of one photon depends only on its frequency (Planck, 1901):

$$\varepsilon = hf = \frac{hc}{\lambda}, \tag{12.2}$$

where

ε	energy of one photon (J)	calculated
h	Planck's constant (J s)	6.626×10^{-34}, Table B.3
$f,\ c,\ \lambda$	as in Eq. (12.1)	

where the second form follows from substituting $f = c/\lambda$ from Eq. (12.1). So **the shorter the wavelength, the more energy each photon carries**:

$$\varepsilon_{0.5\,\mu\text{m}} = \frac{(6.626 \times 10^{-34})(3.00 \times 10^{8})}{0.5 \times 10^{-6}} = 3.98 \times 10^{-19}\ \text{J},$$
$$\varepsilon_{10\,\mu\text{m}} = \frac{(6.626 \times 10^{-34})(3.00 \times 10^{8})}{10 \times 10^{-6}} = 1.99 \times 10^{-20}\ \text{J}.$$

A photon of sunlight carries twenty times the energy of a photon of the Earth's infrared. This is why ultraviolet light, with even shorter wavelengths, can break chemical bonds and burn skin, while infrared only warms.

12.4 The electromagnetic spectrum

Table 12.1 and Figure 12.2 arrange the kinds of radiation by wavelength. Two ranges matter for the atmosphere. The Sun emits mostly between about 0.2 and 4 μm (ultraviolet, visible and near infrared): this is called **shortwave** radiation. The Earth and its atmosphere emit mostly between about 4 and 100 μm (thermal infrared): this is **longwave** radiation. Most of the Sun's ultraviolet never reaches the ground: ozone absorbs it, and that absorption is what warms the stratosphere in the temperature profiles of Part I (§ 5.4).

Table 12.1. The electromagnetic spectrum. All of these are photons; they differ only in wavelength and so in energy per photon, Eq. (12.2).

Type	Wavelength	Energy per photon	Example source
Radio	metres to kilometres	smallest	transmitters
Microwave	1 mm to 1 m	very small	ovens; radar
Infrared	0.75 to 1000 μm	small	the Earth, people, warm objects (peak near 10 μm)
Visible	0.4 to 0.75 μm	moderate	the Sun (peak near 0.5 μm)
Ultraviolet	0.01 to 0.4 μm	large	the Sun; mostly absorbed by ozone
X-ray, gamma	shorter still	largest	medical X-rays; radioactive decay

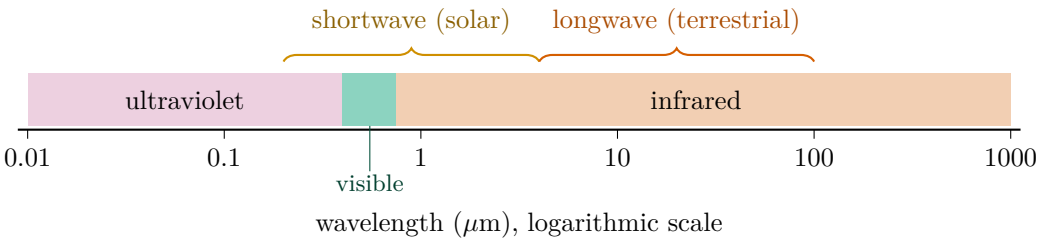


Figure 12.2. The part of the spectrum that matters for the atmosphere. Solar (shortwave) and terrestrial (longwave) radiation meet near 4 μm.

12.5 Why everything above absolute zero radiates

The first law of radiation says that *all objects continually radiate energy.* Why?

Absolute zero (0 K, or −273.15 °C) is the lowest possible temperature, at which the particles of a substance have the least possible motion. The kelvin scale starts there; that is why the radiation laws need temperatures in kelvin (§ 11.2).

All matter (rock, soil, sea, ice, cloud, air, people) is made of atoms that contain negative electrons and positive nuclei. Above absolute zero these particles are always vibrating and colliding, so their electric charges are always being jostled. Moving charges send out electromagnetic waves, just as a radio transmitter drives electrons up and down an aerial. So

$$\text{above 0 K} \;\Rightarrow\; \text{particles move} \;\Rightarrow\; \text{charges are jostled} \;\Rightarrow\; \text{radiation is emitted.}$$

Cold things radiate too, only less. An ice cube at −10 °C gives off about 270 W m^{-2}, and your skin about 500 W m^{-2} (you will be able to check both with Eq. (14.1) in Chapter 14). A thermal camera sees people in the dark for exactly this reason. Ordinary objects look dark only because almost all their emission is infrared, which our eyes cannot see.

12.6 Black bodies

Definition

A **black body** is an ideal object that absorbs *all* the radiation falling on it, at every wavelength, and reflects none. At any given temperature it also *emits* the most radiation that any object at that temperature can emit, at every wavelength.

The last law, *good absorbers are good emitters*, is Kirchhoff's law: at each wavelength, the fraction of incoming radiation a surface absorbs equals the fraction of black-body radiation it emits. That fraction is the **emissivity**, ϵ, between 0 and 1; a black body has $\epsilon = 1$ at all wavelengths (Wallace and Hobbs, 2006, ch. 4).

No real object is a perfect black body, but many come close at some wavelengths. Snow is white (a poor absorber) in visible light but almost black in the thermal infrared, where its emissivity is close to 1; most natural surfaces, water and thick cloud behave nearly as black bodies in the infrared. The Sun is close to a black body over the whole spectrum. This is why the black-body laws work so well here.

12.7 The Planck curve

How much energy does a black body emit at each wavelength? Imagine sorting all the radiation leaving a hot plate into wavelength "bins" and plotting how much energy lands in each bin. The result is a **spectrum**, and for a black body its shape was found by Planck (1901):

$$B_\lambda(T) = \frac{2hc^2}{\lambda^5} \frac{1}{\exp\left(\dfrac{hc}{\lambda k_B T}\right) - 1}. \tag{12.3}$$

where

B_λ	spectral radiance: power per unit area, per unit solid angle, per unit wavelength ($\mathrm{W\,m^{-2}\,sr^{-1}\,m^{-1}}$)	calculated
h, c	as in Eqs. (12.1) and (12.2)	Table B.3
k_B	Boltzmann's constant ($\mathrm{J\,K^{-1}}$)	1.381×10^{-23}, Table B.3
λ	wavelength (m)	the horizontal axis
T	absolute temperature of the body (K)	e.g. 5796 K for the Sun (Question 1)

You will not need to calculate with Eq. (12.3). What matters is the shape it draws (Figure 12.3) and the two laws that come out of it.

- **The curve rises steeply, peaks, then tails off slowly.** At short wavelengths each photon is expensive in energy, Eq. (12.2), and only the most energetic particles can make it, so emission falls off a cliff. At long wavelengths photons are cheap but each carries little, so the curve tails off gradually.
- **The position of the peak** says *what kind* of radiation the body mainly emits. Hotter bodies peak at shorter wavelengths. This is Wien's law (Chapter 13).
- **The area under the curve** is the *total* radiation emitted. Hotter bodies have far larger areas. This is the Stefan–Boltzmann law (Chapter 14).

A geographer's analogy is the storm hydrograph. Its peak tells you *when* discharge was greatest; the area under it tells you the total *volume* of water. The Planck curve is read in the same way, with energy in place of water and wavelength in place of time.

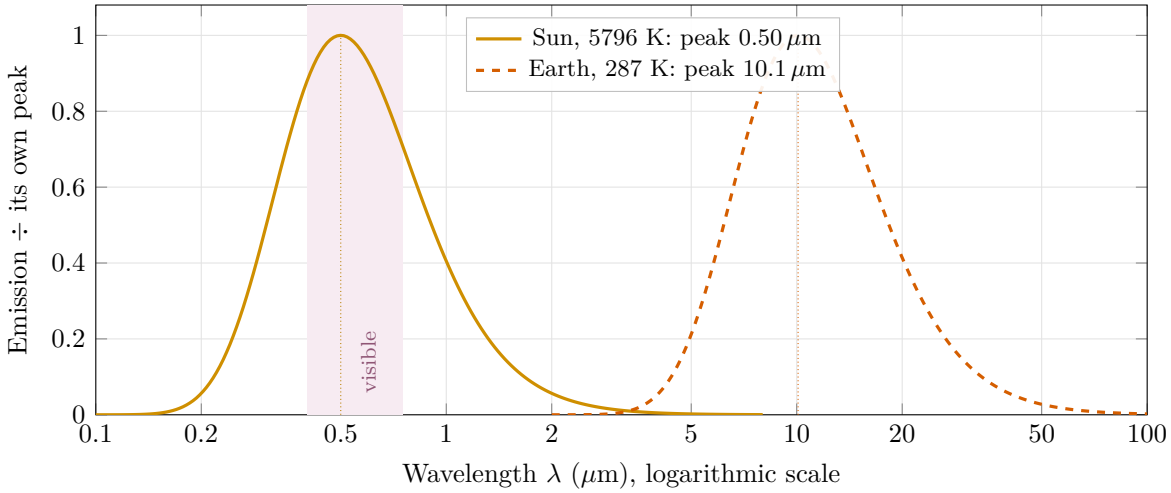


Figure 12.3. Black-body spectra of the Sun and the Earth from Eq. (12.3), each divided by its own peak value so that both can be seen on one graph. The two curves hardly overlap: the Sun's radiation is shortwave and the Earth's longwave. On a common scale the Earth curve would be invisible, because per square metre the Sun emits about 166 000 times more energy (Chapter 14).

Four laws of radiation, and where each is used

1. *All objects continually radiate energy over a range of wavelengths* (§ 12.5; the spread is the Planck curve, Eq. (12.3)).
2. *Hotter objects radiate more total energy per unit area*: the Stefan–Boltzmann law, Eq. (14.1), used in Questions 2 and 4.
3. *Hotter objects radiate more at short wavelengths*: Wien's displacement law, Eq. (13.1), used in Questions 1 and 3.
4. *Good absorbers are good emitters*; the perfect one is a black body (§ 12.6), the assumption behind Questions 3 and 4.

12.8 Questions that come up

A question that comes up

If the air radiates too, why do we not feel it?
We do. On a clear night the ground cools quickly because it radiates to a cold sky; under cloud it stays warmer because the cloud radiates back to it. That downward infrared from cloud and air is part of the greenhouse effect of Chapter 17.

A question that comes up

When does an object start to glow visibly?
When enough of its Planck curve reaches into the visible. For a black body this begins at roughly 800 K, with a dull red glow, as in an electric hob ring. At the Sun's temperature the peak is in the middle of the visible.

12.9 Try it yourself: Practice Set 12

Practice Set 12: waves and photons

Use $c = 3.00 \times 10^{8}\,\mathrm{m\,s^{-1}}$ and $h = 6.626 \times 10^{-34}\,\mathrm{J\,s}$.

12.1 What is the frequency of light with a wavelength of $0.5\,\mu$m?

12.2 What is the energy of one photon of wavelength $10\,\mu$m?

12.3 A microwave oven works at a wavelength of 12.2 cm. What is its frequency?

12.4 How many times more energy does a $0.3\,\mu$m ultraviolet photon carry than a $3\,\mu$m infrared photon?

12.5 Name the part of the spectrum, and say whether it is shortwave or longwave, for radiation of 0.3, 0.55, 2 and $15\,\mu$m.

Answers to Practice Set 12

12.1 $f = c/\lambda = 3.00 \times 10^{8} / 0.5 \times 10^{-6} = 6.0 \times 10^{14}\,\mathrm{Hz}$, Eq. (12.1).

12.2 $\varepsilon = hc/\lambda = (6.626 \times 10^{-34})(3.00 \times 10^{8})/(10 \times 10^{-6}) = 1.99 \times 10^{-20}\,\mathrm{J}$, Eq. (12.2).

12.3 $f = 3.00 \times 10^{8} / 0.122 = 2.46 \times 10^{9}\,\mathrm{Hz}$ (2.46 GHz).

12.4 The energy is inversely proportional to the wavelength, so the ratio is $3/0.3 = 10$ times.

12.5 $0.3\,\mu$m: ultraviolet, shortwave. $0.55\,\mu$m: visible (green), shortwave. $2\,\mu$m: near infrared, shortwave (still part of the solar spectrum). $15\,\mu$m: thermal infrared, longwave.

Chapter 13

Wien's law and the temperature of the Sun

This chapter answers Question 1.

13.1 Hotter bodies peak at shorter wavelengths

Heat a piece of iron in a forge. At first it gives off only invisible infrared; then it glows dull red, then orange, then yellow-white. As it gets hotter, the colour of its glow moves towards shorter wavelengths. Figure 13.1 shows the same thing with Planck curves, Eq. (12.3): as the temperature rises, every curve gets taller, and its peak slides to the left.

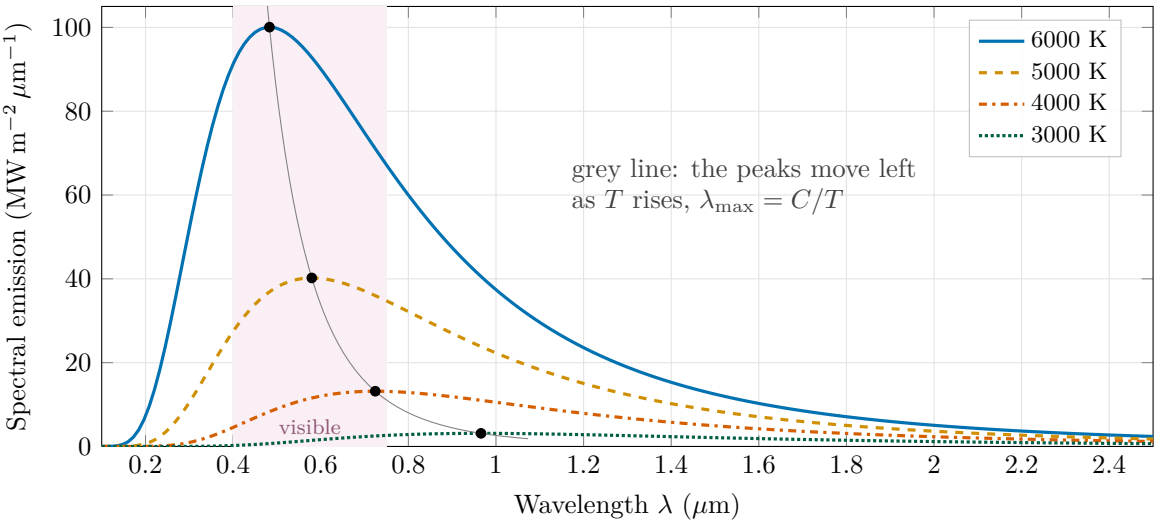


Figure 13.1. Planck curves, Eq. (12.3), for four temperatures, drawn as the power emitted per square metre per micrometre of wavelength. The dots mark the peaks and the grey line joins them: the hotter the body, the shorter the wavelength of its peak (Wien's law, Eq. (13.1)) and the larger the area under its curve (the Stefan–Boltzmann law, Eq. (14.1)).

13.2 Wien's displacement law

The wavelength of the peak is inversely proportional to the temperature:

$$\lambda_{\max} = \frac{C}{T}. \tag{13.1}$$

where

$\lambda_{\max}$	wavelength of peak emission (μm)	read from a spectrum, or calculated
C	Wien's constant (μm K)	2898, rounded
T	absolute temperature of the body (K)	calculated, or given

"Inversely proportional" means that when one doubles the other halves: a body twice as hot peaks at half the wavelength. The constant C is not arbitrary. It follows from finding the peak of the Planck curve, which is done step by step in Appendix B.2.1; the result is $C = hc/(4.965\,k_B) = 2897.8\,\mu\text{m K}$.

To find a temperature from a peak wavelength, multiply both sides of Eq. (13.1) by T and divide both sides by $\lambda_{\max}$:

$$T = \frac{C}{\lambda_{\max}}. \tag{13.2}$$

where

T, C, $\lambda_{\max}$ as in Eq. (13.1)

The units work out: μm K divided by μm leaves K. This is why $\lambda_{\max}$ must be in micrometres when $C = 2898\,\mu\text{m K}$.

13.3 Question 1: the temperature of the Sun

The question

If the wavelength emitted more than any other by the Sun is $0.5\,\mu$m, what is the black-body temperature of the Sun in Celsius? Note the conversion from kelvin to Celsius is $^\circ\text{C} = \text{K} - 273.16$.

What the question asks, in plain words. We know where the Sun's spectrum peaks ($0.5\,\mu$m, in the blue-green part of the visible). Assuming the Sun is a black body, Wien's law tells us what temperature produces a peak there.

Method.

Step 1. Write down the rearranged law, Eq. (13.2): $T = C/\lambda_{\max}$.

Step 2. Substitute $C = 2898\,\mu\text{m K}$ and $\lambda_{\max} = 0.5\,\mu$m:

$$T_{\text{Sun}} = \frac{2898\ \mu\text{m K}}{0.5\ \mu\text{m}} = 5796\ \text{K}.$$

Step 3. Convert to Celsius with the offset the question gives:

$$T_{\text{Sun}} = 5796 - 273.16 = 5522.84\ ^\circ\text{C} \approx 5523\ ^\circ\text{C}.$$

Answer to Question 1

$T_{\text{Sun}} = 2898/0.5 = \mathbf{5796\,K}$, which is $5796 - 273.16 = \mathbf{5522.8}\,^\circ$C (about $5500\,^\circ$C).

Check. Measurements give the Sun an effective temperature of 5772 K (Prša et al., 2016), within half a percent of our 5796 K: the small difference comes from rounding the peak to $0.5\,\mu$m (the exact value for 5772 K is $0.502\,\mu$m). The order of magnitude is also sensible: a body of a few thousand kelvin glows white-hot, and the Sun's peak lies in the visible, which is the light our eyes evolved to use.

The 5796 K found here is used again in Question 2a (§ 14.3) and Question 2c (§ 14.5).

A question that comes up

Why does the answer say 5522.8 °C and not 5522.85 °C?
The question subtracts 273.16; the exact offset between the kelvin and Celsius scales is 273.15 (Tiesinga et al., 2021). Either way the answer rounds to 5523 °C, and at the precision of a peak read as "0.5 μm" the last digits mean nothing. Quote about 5500 °C when you describe it in words.

A question that comes up

If the Sun peaks in blue-green light, why does it look white, or yellow?
Because its spectrum is broad (Figure 12.3): it emits strongly at all visible wavelengths, and the mixture looks white. Near the horizon, the long path through the air scatters blue light out of the beam, so the Sun looks yellow or red. The position of the peak also depends on how the spectrum is drawn (per unit wavelength, as here, or per unit frequency), so "the colour of the peak" is not the colour we see.

13.4 Wien's law in practice

- **Satellites.** A satellite cannot put a thermometer on the ground. It measures radiation in chosen wavelength bands and converts it to temperature with the radiation laws. Wien's law tells the designer which band suits which target: about 10–12 μm for the land, sea and cloud tops (near 300 K and below), and about 4 μm for fires, which are far hotter.
- **Stars.** Astronomers read the temperature of a star from its colour: red stars are cool (around 3000 K) and blue-white stars hot (10 000 K or more).
- **Shortwave and longwave.** Because the Sun is about twenty times hotter than the Earth, Wien's law puts its peak at a wavelength about twenty times shorter. This single fact separates solar from terrestrial radiation (Figure 12.3) and makes the greenhouse effect possible (Chapter 17).

13.5 Try it yourself: Practice Set 13

Practice Set 13: Wien's law

Use $C = 2898\,\mu\text{m K}$ and $°\text{C} = \text{K} - 273.16$.

13.1 The star Sirius has a surface temperature of about 9940 K. At what wavelength does it emit most, and in which part of the spectrum is that?

13.2 Human skin is at about 33 °C. At what wavelength does it emit most?

13.3 A star's spectrum peaks at 1.0 μm. What is its temperature in kelvin and in Celsius?

13.4 A forest fire burns at about 1000 K. Where is its peak? Why do satellites look for fires near 4 μm rather than at 10 μm?

13.5 An ice cube at −10 °C emits radiation. At what wavelength does it emit most?

Answers to Practice Set 13

13.1 $\lambda_{\text{max}} = 2898/9940 = 0.29\,\mu$m: ultraviolet, Eq. (13.1). Sirius looks blue-white.

13.2 $T = 33 + 273.16 = 306.16$ K, so $\lambda_{\text{max}} = 2898/306.16 = 9.47\,\mu$m, in the thermal infrared. Thermal cameras are built to see near this wavelength.

13.3 $T = 2898/1.0 = 2898\,\mathrm{K} = 2624.8\,^{\circ}\mathrm{C}$, Eq. (13.2). A cool, red star.

13.4 $\lambda_{\max} = 2898/1000 = 2.9\,\mu\mathrm{m}$. Near $4\,\mu\mathrm{m}$ a fire emits far more than the cool ground around it (whose peak is near $10\,\mu\mathrm{m}$), so fires stand out clearly; at $10\,\mu\mathrm{m}$ the contrast is much smaller.

13.5 $T = 263.16\,\mathrm{K}$, so $\lambda_{\max} = 2898/263.16 = 11.0\,\mu\mathrm{m}$. Cold objects emit at longer wavelengths than warm ones.

13.6 Wien's law in Python

These functions compute the equations of this chapter, and the book's tests check them against its answers.

```
def wien_temperature(peak_um):
    """Black-body temperature (K) from the wavelength of peak emission (micrometres)."""
    return WIEN_C / peak_um
```

`wien_temperature(0.5)` returns 5796.0.

Chapter 14

The Stefan–Boltzmann law: how much the Sun and the Earth emit

This chapter answers Questions 2a to 2e.

14.1 The total emission of a black body

Wien's law (Chapter 13) told us *where* a spectrum peaks. The Stefan–Boltzmann law tells us *how much* a body emits in total, adding up every wavelength: it is the area under the Planck curve:

$$E = \sigma T^4. \tag{14.1}$$

where

E	total radiation emitted per square metre of surface, the "intensity of radiation" ($\mathrm{W\,m^{-2}}$)	calculated
σ	Stefan–Boltzmann constant ($\mathrm{W\,m^{-2}\ K^{-4}}$)	5.67×10^{-8}, rounded
T	absolute temperature of the surface (K)	from Question 1, or given

The law is derived from the Planck curve in Appendix B.2.2: integrating Eq. (12.3) over all wavelengths gives exactly T^4 times a combination of constants, and that combination is σ.

It applies to a black body. A real surface emits a fraction ϵ (its emissivity, § 12.6) of this: $E = \epsilon\sigma T^4$. For the Sun, and for most natural surfaces in the infrared, ϵ is close to 1, so Eq. (14.1) is a good approximation.

14.2 What a fourth power means

T^4 means $T \times T \times T \times T$. A fourth power grows very fast:

Temperature multiplied by	1.1	1.5	2	3	10	20
Emission multiplied by	1.46	5.06	16	81	10 000	160 000

A body only 10 % hotter emits 46 % more; a body twice as hot emits sixteen times as much. Figure 14.1 shows this on a scale wide enough to hold both the Earth and the Sun.

14.3 Question 2a: the emission of the Sun

The question

Using this equation, calculate how much radiation is emitted at the surface of the Sun.

Method. Use Eq. (14.1) with the Sun's temperature from Question 1, $T = 5796\,\mathrm{K}$ (§ 13.3). Raise to the fourth power by squaring twice; this avoids slips on a calculator:

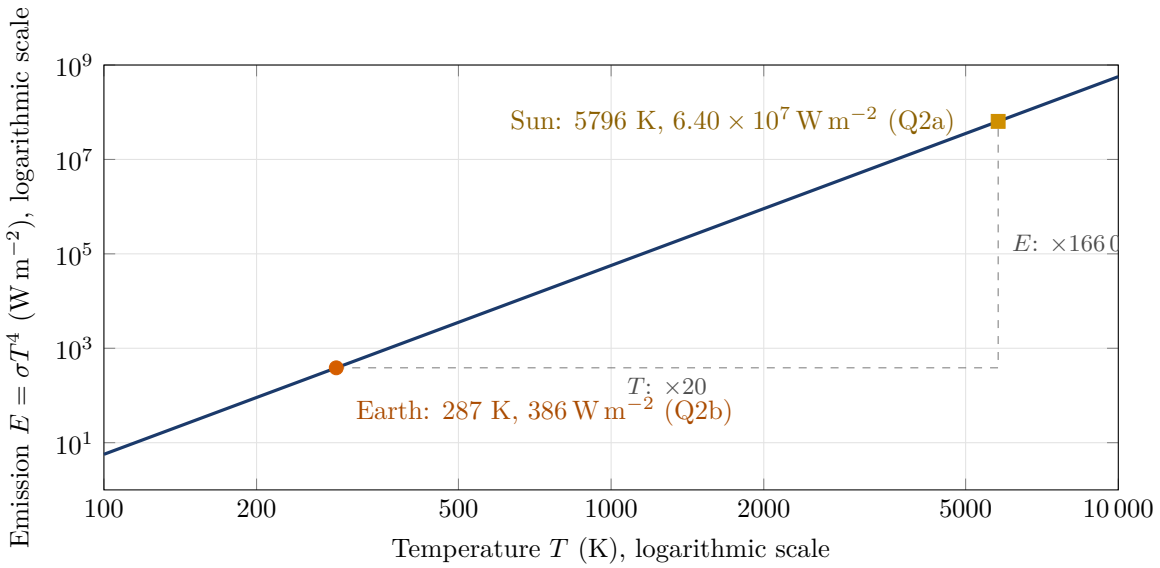


Figure 14.1. The Stefan–Boltzmann law, Eq. (14.1), on logarithmic axes, where a power law becomes a straight line of slope 4. Going from the Earth to the Sun the temperature rises twenty-fold (Question 2c) but the emission rises 166 000-fold (Question 2d).

Step 1. $T^2 = 5796^2 = 3.3594 \times 10^7$.

Step 2. $T^4 = (3.3594 \times 10^7)^2 = 1.1285 \times 10^{15}$.

Step 3. $E_{\text{Sun}} = \sigma T^4 = (5.67 \times 10^{-8})(1.1285 \times 10^{15}) = 6.399 \times 10^7\ \text{W m}^{-2}$.

When multiplying powers of ten, add the exponents: $10^{-8} \times 10^{15} = 10^7$.

Answer to Question 2a

$E_{\text{Sun}} = 5.67 \times 10^{-8} \times 5796^4 = \mathbf{6.40 \times 10^7}\ \text{W m}^{-2}$: about 64 million watts from every square metre of the Sun's surface.

Check. With the measured effective temperature of 5772 K (Prša et al., 2016) the same calculation gives 6.29×10^7 W m^{-2}, within 2 %. Spread over the whole Sun, this emission is the Sun's total power output, and diluted over the distance to the Earth it becomes the solar constant of Question 3c (see the worked example in § 14.8).

14.4 Question 2b: the emission of the Earth

The question

If the Earth has a black-body temperature of 14 °C, what is its intensity of radiation?

Method.

Step 1. Convert to kelvin: $T = 14 + 273.16 = 287.16$ K. The radiation laws need kelvin (§ 11.2).

Step 2. $T^2 = 287.16^2 = 82\,460.9$, and $T^4 = 82\,460.9^2 = 6.7998 \times 10^9$.

Step 3. $E_{\text{Earth}} = (5.67 \times 10^{-8})(6.7998 \times 10^9) = 385.5\ \text{W m}^{-2}$.

Answer to Question 2b

$E_{\text{Earth}} = 5.67 \times 10^{-8} \times 287.16^4 = \mathbf{385.5}\,\text{W m}^{-2}$.

Check. Observed energy budgets put the emission of the Earth's surface at about $396\,\text{W m}^{-2}$ (Trenberth et al., 2009), close to our value (the real surface is a little warmer in places and the fourth power weights warm places heavily). Notice that $385.5\,\text{W m}^{-2}$ is *more* than the Earth receives from the Sun on average (about $240\,\text{W m}^{-2}$ absorbed, Question 4c). That surplus is supplied by the atmosphere radiating back down to the surface: the greenhouse effect of Chapter 17.

Where the bracket goes

Only the temperature is raised to the fourth power: $E = 5.67 \times 10^{-8} \times (287.16)^4$, not $(5.67 \times 10^{-8} \times 287.16)^4$, which would give about 7×10^{-20}.

14.5 Question 2c: how many times hotter is the Sun?

The question

How many times warmer is the Sun than the Earth (solar temperature in kelvin / Earth temperature in kelvin)?

Method. Divide the two temperatures, both in kelvin, from Questions 1 and 2b:

$$\frac{T_{\text{Sun}}}{T_{\text{Earth}}} = \frac{5796\ \text{K}}{287.16\ \text{K}} = 20.18.$$

Answer to Question 2c

The Sun is $5796/287.16 \approx$ **20 times** hotter than the Earth (20.18), measured on the kelvin scale.

Kelvin, not Celsius

Ratios of temperatures only make sense on the kelvin scale, which starts at absolute zero. On the Celsius scale they are meaningless: $20\,^\circ\text{C}$ is not "twice as hot" as $10\,^\circ\text{C}$ (in kelvin it is $293.16/283.16 = 1.035$ times as hot), and nothing is "-2 times" as hot as $-10\,^\circ\text{C}$.

14.6 Question 2d: how many times more does the Sun emit?

The question

How many more times is the Sun's radiation more intense than the Earth's?

Method. Divide the emissions from Questions 2a and 2b:

$$\frac{E_{\text{Sun}}}{E_{\text{Earth}}} = \frac{6.399 \times 10^7\ \text{W m}^{-2}}{385.5\ \text{W m}^{-2}} = 1.660 \times 10^5.$$

Answer to Question 2d

Each square metre of the Sun emits about **166 000 times** as much radiation as a square metre of the Earth.

14.7 Question 2e: why the two ratios are so different

The question

How do you explain the difference between your answer to Question 2c and 2d?

The idea. Write Eq. (14.1) for both bodies and divide. The constant σ cancels, and only the ratio of temperatures remains, raised to the fourth power:

$$\frac{E_{\text{Sun}}}{E_{\text{Earth}}} = \frac{\sigma T_{\text{Sun}}^4}{\sigma T_{\text{Earth}}^4} = \left(\frac{T_{\text{Sun}}}{T_{\text{Earth}}}\right)^4. \tag{14.2}$$

where

E_{Sun}, E_{Earth}	emissions per square metre (W m^{-2})	Questions 2a and 2b
T_{Sun}, T_{Earth}	temperatures (K)	5796 and 287.16

So the emission ratio is the temperature ratio of Question 2c raised to the fourth power: $20.18^4 = 20.18 \times 20.18 \times 20.18 \times 20.18 = 1.660 \times 10^5$, exactly the answer to Question 2d.

Answer to Question 2e

Emission depends on the **fourth power** of temperature, Eq. (14.1). The Sun is about 20 times hotter, so it emits about $20^4 = 160\,000$ times more per square metre (precisely $20.18^4 = 166\,000$). A modest difference in temperature becomes an enormous difference in emission.

A power law, not an exponential

If T doubles, E becomes 16 times greater, but that does not make the change exponential. In an *exponential* relationship, such as $E \propto e^{kT}$, the temperature sits in the exponent and every equal step in T multiplies E by the same factor. In the Stefan–Boltzmann law the exponent is fixed at 4 and the temperature is the base: this is a **power law**, $E \propto T^4$. An exponential eventually outgrows any power law, and on the log–log axes of Figure 14.1 a power law is a straight line while an exponential is not. Say "E is proportional to the fourth power of T".

14.8 From the Sun's surface to the Earth

Questions 2a and 3c are linked. The Sun's surface emits σT^4 per square metre; this energy spreads out over ever larger spheres as it travels, so by the time it reaches the Earth it has been diluted to the solar constant S.

Worked example: the solar constant from the Sun's temperature

The Sun's radius is $R_\odot = 6.957 \times 10^8$ m and the Earth–Sun distance is $d = 1.496 \times 10^{11}$ m (Prša et al., 2016). The Sun's total power is its emission per square metre times its surface area $4\pi R_\odot^2$:

$$P = \sigma T^4 \times 4\pi R_\odot^2 = (6.399 \times 10^7) \times 4\pi (6.957 \times 10^8)^2 = 3.89 \times 10^{26}\ \text{W}.$$

At the Earth this power is spread over a sphere of radius d, so each square metre facing the Sun receives

$$S = \frac{P}{4\pi d^2} = \frac{3.89 \times 10^{26}}{4\pi(1.496 \times 10^{11})^2} = 1384 \text{ W m}^{-2}.$$

With the measured 5772 K the same steps give 1361 W m^{-2}, the value measured by satellites (Kopp and Lean, 2011). The 1368 W m^{-2} used in this part is an older measurement of the same quantity. The dilution with distance, $S \propto 1/d^2$, is the inverse-square law used for the planets in Chapter 16, Eq. (16.1).

14.9 Questions that come up

A question that comes up

Question 2b calls 14 °C the Earth's "black-body temperature". Later the Earth's black-body temperature is 255 K (−18 °C). Which is right?
They are two different things. 14 °C (287 K) is the average temperature of the Earth's *surface*, and Question 2b treats that surface as a black body. 255 K is the temperature the *planet as a whole* must have to send back to space as much energy as it absorbs (Questions 3c and 4d). The 33 K difference between them is the greenhouse effect (§ 17.7).

A question that comes up

My calculator shows 6.3988E7. How do I write that?
"E7" means "times ten to the power 7": 6.3988×10^7 W m^{-2}, or 63 988 000 W m^{-2}. Round to three or four significant figures, 6.40×10^7 W m^{-2}.

14.10 Try it yourself: Practice Set 14

Practice Set 14: the Stefan–Boltzmann law

Use $\sigma = 5.67 \times 10^{-8}$ W m^{-2} K^{-4} and °C = K − 273.16.

14.1 How much does a black body at 0 °C emit?

14.2 By what percentage does the emission rise if the temperature rises by 10 %?

14.3 What temperature must a black body have to emit 1000 W m^{-2}? Give kelvin and Celsius.

14.4 The Earth's surface warms from 287.16 K to 288.16 K. By how much does its emission increase?

14.5 Using the worked example in § 14.8, explain in one sentence why the Earth receives only about 1368 W m^{-2} from a Sun that emits 6.4×10^7 W m^{-2}.

Answers to Practice Set 14

14.1 $E = 5.67 \times 10^{-8} \times 273.16^4 = 315.7$ W m^{-2}, Eq. (14.1).

14.2 $1.1^4 = 1.464$, so the emission rises by 46 %, Eq. (14.2).

14.3 $T = (E/\sigma)^{1/4} = (1000/5.67 \times 10^{-8})^{1/4} = 364.4$ K = 91.3 °C. The rearrangement is Eq. (17.3).

14.4 $5.67 \times 10^{-8} \times (288.16^4 - 287.16^4) = 5.40$ W m^{-2}. One kelvin of warming adds about

$5.4\,\mathrm{W\,m^{-2}}$ of emission; this is why warming is a strong brake on further warming.

14.5 The Sun's energy spreads over a sphere whose area grows with the square of the distance; at 150 million kilometres each square metre receives only $(R_\odot/d)^2 = 1/46\,000$ of what each square metre of the Sun emits.

14.11 Emission in Python

These functions compute the equations of this chapter, and the book's tests check them against its answers.

```
def wien_peak(t_k):
    """Wavelength of peak emission (micrometres) at temperature t_k."""
    return WIEN_C / t_k
```

```
def emission(t_k):
    """Stefan-Boltzmann law: emission (W per square metre) at temperature t_k."""
    return SIGMA * t_k ** 4
```

`emission(5796)` returns about 6.399×10^7 and `emission(287.16)` about 385.5.

Chapter 15

Plotting three planetary spectra

This chapter answers the graph of Question 3 and Questions 3a and 3b.

15.1 The data

Table 15.1 gives the spectral emission of three black bodies, labelled Data A, B and C, at fourteen wavelengths from 4 to 30 μm Ėach column is one planet; the task is to plot all three on one graph, read off where each peaks, and use Wien's law to find each planet's temperature.

Table 15.1. The data of Question 3. The largest value in each column is in bold. The same table, with the values computed from the Planck curve, is in Appendix B.3.

Wavelength (μm)	Data A	Data B	Data C
4	26.199	1.3338	0.11992
6	386.28	53.056	10.648
8	970.65	218.86	65.619
10	1313.3	398.14	151.8
12	**1363.5**	502.49	224.7
14	1248.6	**527.86**	264.23
16	1074.3	502.41	**273.39**
18	896.25	452.58	262.51
20	737.1	395.22	241.03
22	603.1	339.25	215.38
24	493.45	288.62	189.36
26	404.91	244.61	164.92
28	333.76	207.16	142.9
30	276.58	175.67	123.56

A note on the units of the data

The columns are headed "Irradiance, W m^{-2}". Strictly these are *spectral* values: emission per square metre *per micrometre of wavelength* (W m^{-2} μm^{-1}), on an arbitrary scale. Recomputing them shows that each column is a Planck curve, Eq. (12.3), multiplied by the same constant (about 316) for all three (Appendix B.3). The units do not affect any answer, because the questions only use *where* each curve peaks and how the curves compare.

15.2 How to plot the graph

Step 1. Axes. Wavelength (μm) along the bottom, from 0 to 30. Emission up the side, from 0 to at least 1400, since the largest value is 1363.5.

Step 2. Scale. Choose round steps that use most of the paper: for example 2 μm per major division across and 100 or 200 per division up.

Step 3. Plot each data set with its own symbol (for example crosses for A, circles for B and triangles for C), so that the graph still reads correctly in black and white.

Step 4. Join the points with a smooth curve, not with straight lines. The underlying spectrum is smooth (§ 12.7); straight segments would put the peak exactly on a data point, which hides the fact that the true peak may lie between two points.

Step 5. Label each curve (A, B, C), the axes with quantity and unit, and give the graph a title.

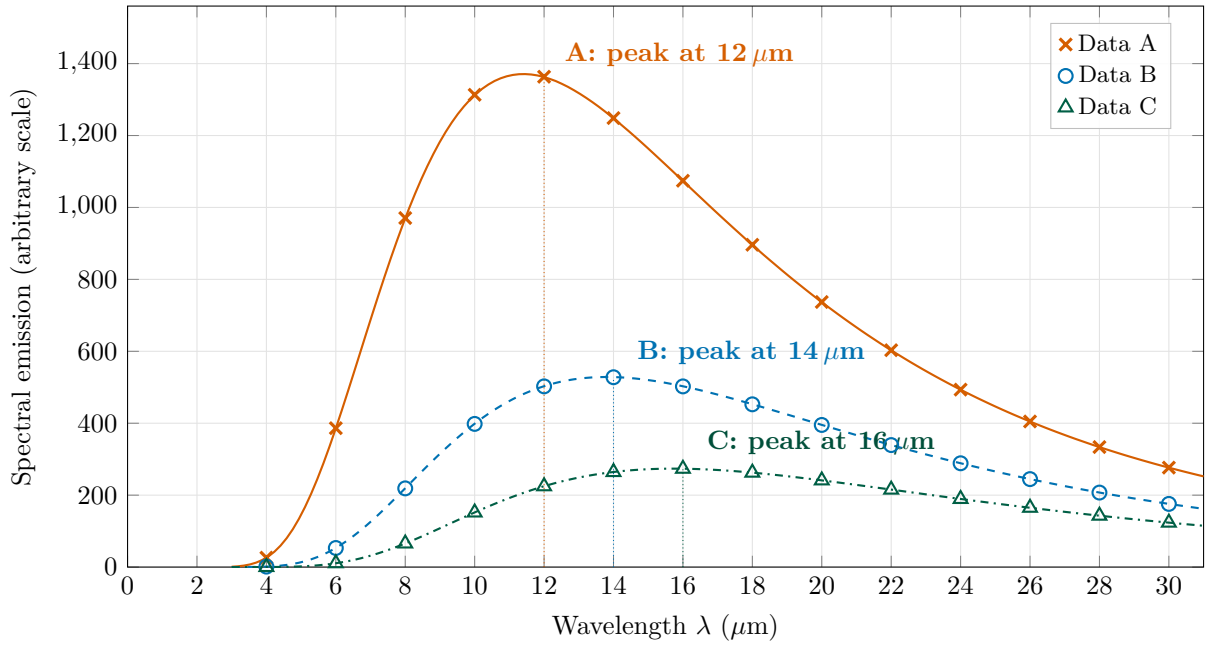


Figure 15.1. The graph of Question 3. Symbols are the values of the table; the smooth lines are the Planck curves that pass through them (254.1, 210.0 and 184.1 K; Appendix B.3). The dotted lines mark the highest data point of each set, which is the answer to Question 3a.

15.3 Question 3a: the wavelength of the peak

The question

At what wavelength does the maximum intensity occur for each data set?

Method. Find the largest number in each column of Table 15.1 (in bold), or the highest point of each curve in Figure 15.1, and read its wavelength.

Answer to Question 3a

Data A: $\lambda_{\max} = \mathbf{12}\,\mu$m (1363.5). Data B: $\lambda_{\max} = \mathbf{14}\,\mu$m (527.86). Data C: $\lambda_{\max} = \mathbf{16}\,\mu$m (273.39).

Check. The curve with the highest peak (A) peaks furthest to the left, and the lowest (C) furthest to the right. That is what Wien's law and the Stefan–Boltzmann law together predict (Figure 13.1): the hottest body is both the brightest and the shortest in wavelength. All three peaks lie in the thermal infrared, as they must for planets far cooler than the Sun.

15.4 Question 3b: the temperatures of the three planets

The question

Using Wien's law, calculate the black-body temperature for the three planets.

Method. Use the rearranged Wien's law, Eq. (13.2), $T = C/\lambda_{\max}$, with $C = 2898\,\mu\text{m K}$ and the peaks from Question 3a:

$$T_A = \frac{2898}{12} = 241.5 \text{ K},$$
$$T_B = \frac{2898}{14} = 207.0 \text{ K},$$
$$T_C = \frac{2898}{16} = 181.1 \text{ K}.$$

Answer to Question 3b

$T_A = \mathbf{241.5\,K}$ ($-31.7\,^\circ$C), $T_B = \mathbf{207.0\,K}$ ($-66.2\,^\circ$C), $T_C = \mathbf{181.1\,K}$ ($-92.0\,^\circ$C).

Check. The order is right: A is the warmest and C the coldest, matching the heights of the curves. The values are cold, which is reasonable for planets heated by a distant Sun; Chapter 16 identifies which planet each curve belongs to.

Why these temperatures are a little too low

The data are given only every $2\,\mu$m, so the highest *data point* is not exactly the *peak* of the curve. Fitting the Planck curve to each column (Appendix B.3) shows that the true peaks are at 11.4, 13.8 and 15.7 μm, and the true temperatures are **254.1, 210.0 and 184.1 K**. Reading the nearest point puts every peak slightly too far to the right, and so every temperature slightly too low; the error is largest for A (12 instead of 11.4 μm, so 241.5 instead of 254 K). The answers above are the right answers to the question as set; the true values matter in Chapter 16, where they match the calculated planetary temperatures almost exactly.

A question that comes up

Could I find the peak more accurately from the data?
Partly. Fitting a parabola through the three highest points of a column and taking its top moves the A peak from 12 to 11.6 μm (249.7 K), closer to the truth. It does much less for B and C, because a Planck curve is lopsided (steep on the short-wave side, with a long tail) while a parabola is symmetric. The reliable way is to fit the Planck curve itself, as in Appendix B.3, or to take readings closer together near the peak.

15.5 What the graph shows

- **The curves never cross.** At every wavelength the warmer body emits more. This is a property of all Planck curves: heating a black body raises its emission at every wavelength.
- **The area under each curve** is the total emission, which grows as T^4, Eq. (14.1). With the true temperatures, A emits $(254.1/184.1)^4 = 3.6$ times as much in total as C.
- **The gap between the curves is largest on the short-wave side.** At 10 μm A emits 8.7 times as much as C; at 30 μm only 2.2 times. Short wavelengths are the most sensitive to temperature, which is why the peak moves (Wien's law) and why satellites measuring fires use short infrared bands (§ 13.4).

15.6 Try it yourself: Practice Set 15

Practice Set 15: reading spectra

15.1 If you had readings every 0.2 μm near the top of curve A, you would find its peak at 11.4 μm. What temperature does that give?

15.2 Using Table 15.1, how many times more does A emit than C at 10 μm? At 30 μm? Why are the two ratios different?

15.3 A fourth planet's spectrum peaks near 25 μm. What is its temperature, and which planet of Table 16.1 might it be?

15.4 Explain in one sentence why two Planck curves for different temperatures can never cross.

Answers to Practice Set 15

15.1 $T = 2898/11.4 = 254.2$ K, Eq. (13.2). That is the Earth's calculated black-body temperature (Question 3c).

15.2 $1313.3/151.8 = 8.7$ at 10 μm; $276.58/123.56 = 2.2$ at 30 μm. Emission on the short-wave side of the peak depends far more strongly on temperature than on the long-wave tail, Eq. (12.3).

15.3 $T = 2898/25 = 116$ K: Jupiter, whose calculated black-body temperature is 110 K (Question 3c).

15.4 Raising T in Eq. (12.3) makes the exponential in the denominator smaller at every wavelength, so the emission rises at every wavelength and the warmer curve lies above the cooler one everywhere.

Chapter 16

The temperatures of the planets

This chapter answers Questions 3c, 3d and 3e.

16.1 A planet in balance

A planet is heated by the Sun and cools by radiating to space. If it absorbed more than it emitted it would warm up; warming raises its emission as T^4, Eq. (14.1), until the two are equal. A planet that has settled is therefore **in radiative balance**:

$$\text{solar energy absorbed} = \text{infrared energy emitted}.$$

The temperature at which this happens is the planet's **equilibrium** or **black-body temperature**. To calculate it we need three things: how much sunlight arrives (the solar constant S), how much of it is reflected (the albedo a), and the geometry of a sphere.

16.2 Sunlight at each planet: the inverse-square law

Sunlight spreads out over ever larger spheres as it travels from the Sun (§ 14.8). The area of a sphere grows with the square of its radius, so the energy reaching each square metre falls with the square of the distance:

$$S = \frac{S_E}{d^2}. \tag{16.1}$$

where

S	solar constant at the planet: sunlight per square metre facing the Sun, at the top of the atmosphere ($\mathrm{W\,m^{-2}}$)	calculated; listed in the table
S_E	solar constant at the Earth ($\mathrm{W\,m^{-2}}$)	1368, an older measurement
d	planet's distance from the Sun, in astronomical units (AU; Earth–Sun = 1)	listed in the table

The values in the table follow this law exactly. For Mars, $d = 1.52\,\mathrm{AU}$, so $S = 1368/1.52^2 = 1368/2.310 = 592\,\mathrm{W\,m^{-2}}$: at one and a half times the distance, Mars receives only 43 % of the Earth's sunlight. For Mercury, $1368/0.39^2 = 8994\,\mathrm{W\,m^{-2}}$, six and a half times the Earth's.

16.3 Albedo

Definition

The **albedo** a of a surface or a planet is the fraction of the sunlight falling on it that is reflected, from 0 (absorbs everything) to 1 (reflects everything). The fraction absorbed is $1 - a$.

Fresh snow and thick cloud reflect most sunlight (albedo up to 0.8–0.9); forests, soils and especially open water reflect little (water below about 0.1 with the Sun high in the sky) (Ahrens and Henson, 2022, ch. 2). A planet's overall albedo, called its *Bond albedo*, is the average over its whole sunlit side, cloud included: about 0.3 for the Earth (Forster et al., 2021).

16.4 The black-body temperature of a planet

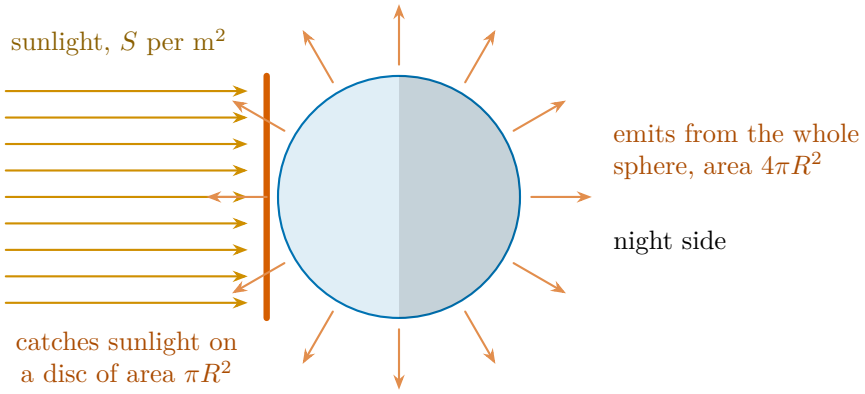


Figure 16.1. Why the solar constant is divided by four. A planet intercepts sunlight over a disc of area πR^2 (its shadow) but radiates from its whole surface, $4\pi R^2$, day and night side alike. Averaged over the sphere, each square metre receives $S/4$.

Set the energy absorbed equal to the energy emitted, for a planet of radius R (Figure 16.1):

$$\text{absorbed} = S\,(1-a) \times \pi R^2, \qquad \text{(sunlight on the disc, minus what is reflected)}$$
$$\text{emitted} = \sigma T^4 \times 4\pi R^2. \qquad \text{(Stefan–Boltzmann law, Eq. (14.1), over the whole sphere)}$$

Setting them equal and cancelling πR^2 from both sides, the size of the planet drops out:

$$S\,(1-a) = 4\sigma T^4 \quad \Longrightarrow \quad T^4 = \frac{S\,(1-a)}{4\sigma},$$

and taking the fourth root of both sides gives the formula in the hint to Question 3c:

$$T = \left[\frac{S\,(1-a)}{4\sigma}\right]^{1/4}. \tag{16.2}$$

where

T	black-body (equilibrium) temperature of the planet (K)	calculated
S	solar constant at the planet (W m^{-2})	in the table; Eq. (16.1)
a	planet's albedo (0 to 1)	in the table
σ	Stefan–Boltzmann constant	5.67×10^{-8} W m^{-2} K^{-4}
4	ratio of the sphere's area to the disc's, Figure 16.1	geometry

A fourth root is a square root taken twice; on a calculator use $x^{0.25}$ or press $\sqrt{\ }$ twice.

16.5 Question 3c: calculating the temperatures

The question

Using the table below, calculate the black-body temperatures T for the following planets (hint: $T = [S(1-a)/(4\sigma)]^{1/4}$).

Worked example: the Earth, step by step

Step 1. Absorbed sunlight per square metre of disc: $S(1-a) = 1368 \times (1-0.3) = 1368 \times 0.7 = 957.6\,\mathrm{W\,m^{-2}}$.

Step 2. Divide by $4\sigma = 4 \times 5.67 \times 10^{-8} = 2.268 \times 10^{-7}$: $957.6 / 2.268 \times 10^{-7} = 4.2222 \times 10^{9}\,\mathrm{K^4}$.

Step 3. Square root: $\sqrt{4.2222 \times 10^{9}} = 64\,979$.

Step 4. Square root again: $\sqrt{64\,979} = 254.9\,\mathrm{K}$.

So $T_{\mathrm{Earth}} = 254.9\,\mathrm{K}$, or $254.9 - 273.16 = -18.3\,^{\circ}\mathrm{C}$.

The same four steps for every planet give Table 16.1.

Table 16.1. Question 3c completed. The first four columns and the last but one are given; $S(1-a)$ and T are calculated with Eq. (16.2); the last column is the actual minus the calculated temperature.

Planet	d (AU)	S ($\mathrm{W\,m^{-2}}$)	a	$S(1-a)$	T calc. (K)	actual (K)	difference
Mercury	0.39	8994	0.06	8454.4	**439.4**	440	+0.6
Venus	0.72	2639	0.90	263.9	**184.7**	737	+552.3
Earth	1.00	1368	0.30	957.6	**254.9**	288	+33.1
Mars	1.52	592	0.25	444.0	**210.3**	210	−0.3
Jupiter	5.18	51	0.34	33.7	**110.4**	152	+41.6

Answer to Question 3c

Mercury **439 K**, Venus **185 K**, Earth **255 K**, Mars **210 K**, Jupiter **110 K** (to the nearest kelvin; Table 16.1).

Check. The temperatures fall with distance from the Sun, as expected, with one exception: Venus is calculated colder than the Earth although it is closer to the Sun. Question 3e asks why. NASA's fact sheets, which use slightly different solar constants and albedos, give black-body temperatures of 439.6 K for Mercury, 254.0 K for the Earth, 209.8 K for Mars and 109.9 K for Jupiter (Williams, 2024), all within a kelvin or so of ours.

16.6 Question 3d: which spectrum belongs to which planet?

The question

Using the temperatures from the table above, link the data sets you plotted previously with the planet.

Method. Compare the temperatures from the spectra (Question 3b, § 15.4) with the calculated temperatures in Table 16.1, and pair each data set with the closest planet.

	From the peak (Q3b)	From the full curve (§ 15.4)	Closest planet (Q3c)
Data A	241.5 K	254.1 K	Earth, 254.9 K
Data B	207.0 K	210.0 K	Mars, 210.3 K
Data C	181.1 K	184.1 K	Venus, 184.7 K

Answer to Question 3d

Data A: **Earth**. Data B: **Mars**. Data C: **Venus**.

Check. No other pairing works: Mercury (439 K) would peak near 6.6 μm and Jupiter (110 K) near 26 μm, Eq. (13.1), both outside the three peaks we found. Fitted to the full curves, the three data sets match the calculated temperatures to within one kelvin: the spectra of Table 15.1 correspond to exactly these planetary temperatures.

16.7 Question 3e: why the real temperatures differ

The question

Using the table you constructed, explain: (i) why Mars is theoretically colder than Earth; (ii) why Venus is theoretically colder than Earth; (iii) what the actual surface temperature of Venus is, how you explain the difference between the theoretical and actual temperature, and which process could be causing this; (iv) of all these planets, Mercury is the only one that has an actual average temperature which is the same as its black-body temperature. Why is that?

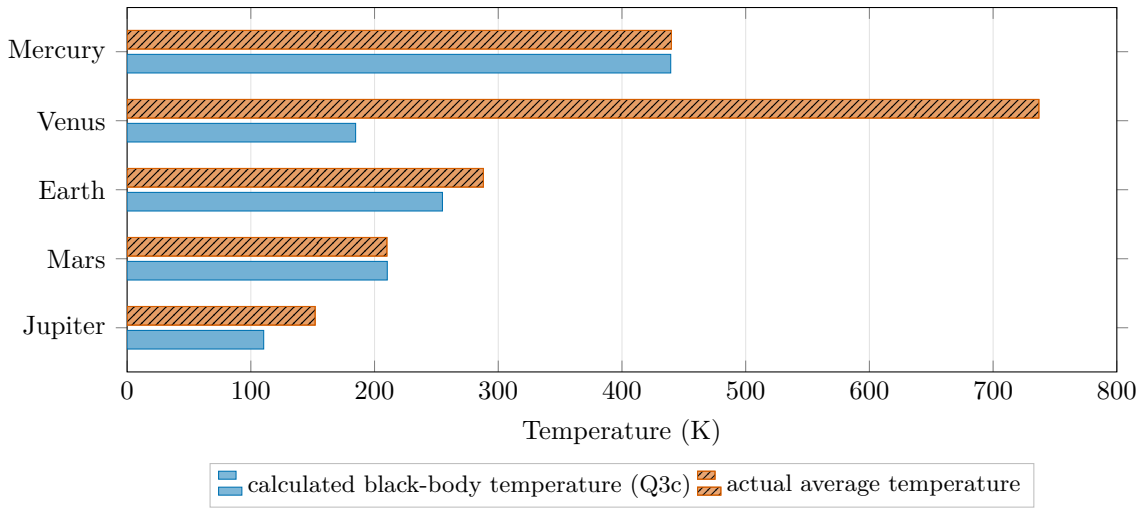


Figure 16.2. Calculated and actual temperatures from Table 16.1. Where the bars match (Mercury, Mars) there is little or no atmosphere; where the actual temperature is higher, an atmosphere traps heat (Venus, Earth) or the planet has an internal source of heat (Jupiter).

(i) Why Mars is theoretically colder than the Earth

Mars is 1.52 times as far from the Sun, so by the inverse-square law, Eq. (16.1), it receives only $1/1.52^2 = 43\,\%$ of the Earth's sunlight: 592 against 1368 W m^{-2}. Its albedo is a little lower (0.25 against 0.30), so it absorbs a slightly larger fraction, but that cannot make up for receiving less than half as much. It absorbs 444 W m^{-2} (per square metre of disc) against the Earth's 957.6, and T scales as the fourth root of that: $(444/957.6)^{1/4} = 0.825$, so $0.825 \times 254.9 = 210$ K.

(ii) Why Venus is theoretically colder than the Earth

Venus is closer to the Sun and receives almost twice the Earth's sunlight (2639 W m^{-2}). But it is wrapped in a permanent, bright cloud deck that reflects most of that light: with an albedo of 0.9

it absorbs only 10 %, which is 263.9 W m^{-2}, less than a third of what the Earth absorbs and even less than Mars. **Albedo, not distance, controls its black-body temperature.**

(iii) The actual surface temperature of Venus

The actual average surface temperature of Venus is about **737 K (464 °C)** (Williams, 2024), hot enough to melt lead, and almost the same by day and by night. That is 552 K hotter than its black-body temperature. The process responsible is the **greenhouse effect**. Venus has a very thick atmosphere (a surface pressure of 92 bar, about ninety times the Earth's), made almost entirely of carbon dioxide, which absorbs the infrared radiation emitted by the surface (Chapter 17). Only the cold upper atmosphere can radiate freely to space, and it does so at roughly the black-body temperature; to push the energy up through such a thick, opaque atmosphere, the surface must be extremely hot (Pierrehumbert, 2011).

(iv) Why Mercury's actual and black-body temperatures agree

Mercury has practically no atmosphere (a surface pressure of less than 10^{-14} bar; Williams, 2024). With no gases to absorb its infrared emission, nothing traps heat, so the surface radiates straight to space exactly as the black-body calculation assumes. Its *average* temperature therefore matches: 440 against 439 K. (The average hides a huge range: with no atmosphere to carry heat around, the sunlit side reaches 590–725 K and the night side is far colder.)

Answer to Question 3e

(i) Mars is further from the Sun (1.52 AU) and receives only 43 % of the Earth's sunlight (inverse-square law), so it absorbs less and is colder, despite its slightly lower albedo.

(ii) Venus has a very high albedo (0.9 in the table): its bright clouds reflect most sunlight, so it absorbs less than a third of what the Earth absorbs, although it is closer to the Sun.

(iii) About **737 K (464 °C)**, some 550 K above its black-body temperature. Its dense carbon dioxide atmosphere produces an intense **greenhouse effect**: the surface emission is absorbed by the atmosphere, and the surface must become very hot before the planet can emit to space as much as it absorbs. The black-body calculation leaves the atmosphere out.

(iv) Mercury has almost no atmosphere and so no greenhouse effect: it absorbs and re-radiates directly to space, exactly as Eq. (16.2) assumes.

Three points on Question 3

1. **Mars matches too.** In the table Mars also matches (210 against 210 K). Mars has a very thin atmosphere (about 6 mb, under 1 % of the Earth's) and so only a small greenhouse effect: its measured average is about 214 K (Williams, 2024), a few kelvin above its black-body temperature. Thin or absent atmospheres give a match; thick ones do not.
2. **Venus's albedo.** The 0.9 in the table is high: the measured Bond albedo is 0.77 (Williams, 2024), which gives a black-body temperature of 227 K. Venus is then still colder than the Earth in theory, and the answers to (ii) and (iii) are unchanged.
3. **"Runaway heating".** Venus today is not heating up: it is in balance, absorbing as much as it emits, at a very high surface temperature. The *runaway greenhouse* describes how Venus probably reached that state long ago, as rising temperatures evaporated any oceans and the water vapour trapped still more heat (Kasting, 1988). Use "an intense greenhouse

effect" for the present.

A question that comes up

Why is Jupiter 42 K warmer than its black-body temperature, if it is so far from the Sun?
Because the Sun is not its only source of heat. Jupiter still releases heat left over from its formation and from slow contraction, and Voyager measurements showed that it emits about 1.7 times as much energy as it absorbs from the Sun (Hanel et al., 1981). Equation (16.2) assumes the Sun is the only heat source, so it underestimates Jupiter. The Earth has an internal source too, but it is tiny: the flow of heat from the Earth's interior averages under 0.1 $\mathrm{W\,m^{-2}}$ (Davies and Davies, 2010), less than one two-thousandth of the 238.7 $\mathrm{W\,m^{-2}}$ of sunlight absorbed (Question 4c).

A question that comes up

The Earth is 33 K warmer than its black-body temperature. Is that also the greenhouse effect?
Yes, and Question 4 measures it directly (§ 17.7).

16.8 Try it yourself: Practice Set 16

Practice Set 16: planetary temperatures

Use $\sigma = 5.67 \times 10^{-8}\,\mathrm{W\,m^{-2}\ K^{-4}}$ and $S_E = 1368\,\mathrm{W\,m^{-2}}$.

16.1 Saturn is 9.54 AU from the Sun and has an albedo of 0.34. Find its solar constant and its black-body temperature.

16.2 What would the Earth's black-body temperature be with an albedo of 0.25? Of 0.35?

16.3 The Moon is at the same distance from the Sun as the Earth and has an albedo of about 0.11. What is its black-body temperature?

16.4 What albedo would give the Earth a black-body temperature of 288 K, its actual average? What does your answer tell you?

16.5 If the Earth were moved to half its present distance from the Sun, keeping its albedo of 0.3, what would its black-body temperature be? Find it two ways.

Answers to Practice Set 16

16.1 $S = 1368/9.54^2 = 15.0\,\mathrm{W\,m^{-2}}$, Eq. (16.1); $T = [15.0 \times 0.66/(4 \times 5.67 \times 10^{-8})]^{1/4} = 81.3\,\mathrm{K}$, Eq. (16.2).

16.2 $a = 0.25$: 259.3 K. $a = 0.35$: 250.2 K. A change of 0.1 in albedo changes the temperature by about 9 K.

16.3 $T = [1368 \times 0.89/(2.268 \times 10^{-7})]^{1/4} = 270.7\,\mathrm{K}$. With no atmosphere, the Moon's average is close to this, but its days are scorching and its nights extremely cold (§ 17.8).

16.4 From Eq. (16.2), $a = 1 - 4\sigma T^4/S = 1 - (2.268 \times 10^{-7} \times 288^4)/1368 = -0.14$. A negative albedo is impossible (no planet can absorb more than all the light falling on it). So no albedo can explain the Earth's 288 K: something other than sunlight absorption, the greenhouse effect, must be keeping the surface warmer.

16.5 At half the distance $S = 4 \times 1368 = 5472\,\mathrm{W\,m^{-2}}$, so $T = [5472 \times 0.7/(2.268 \times 10^{-7})]^{1/4} = 360.5\,\mathrm{K}$. Second way: S is four times larger and $T \propto S^{1/4}$, so T is $4^{1/4} = \sqrt{2} = 1.414$ times larger: $1.414 \times 254.9 = 360.5\,\mathrm{K}$.

16.9 Planetary temperatures in Python

These functions compute the equations of this chapter, and the book's tests check them against its answers.

```
def solar_constant(d_au):
    """Inverse-square law: sunlight at d_au from the Sun (W per square metre)."""
    return S_EARTH / d_au ** 2
```

```
def planet_temperature(s_wm2, albedo):
    """Black-body temperature (K) of a planet: incoming sunlight balances its emission."""
    return (s_wm2 * (1 - albedo) / (4 * SIGMA)) ** 0.25
```

`solar_constant(1.52)` returns about 592, the sunlight reaching Mars.

Chapter 17

The Earth's energy balance and the greenhouse effect

This chapter answers Questions 4a to 4g.

17.1 What Question 4 sets out to do

If the Earth had no atmosphere there would be no greenhouse effect, and the planet would be colder. Question 4 estimates *how much* colder, by working out the temperature an airless Earth would need to send back to space all the sunlight it absorbs, and comparing it with the temperature actually observed. It is the calculation of Question 3c for the Earth (§ 16.5), taken one step at a time, with the energy flows made explicit (Figure 17.1).

17.2 Question 4a: how much is absorbed?

The question

If the global albedo of the Earth is 30 %, how much solar radiation is absorbed by the Earth?

Whatever is not reflected is absorbed (§ 16.3). As a fraction of the incoming sunlight,

$$\text{fraction absorbed} = 1 - a, \qquad \text{fraction reflected} = a. \tag{17.1}$$

where

a	albedo, as a fraction	0.30 (30 %), given

Answer to Question 4a

$1 - 0.30 = 0.70$: the Earth absorbs **70 %** of the solar radiation it receives.

17.3 Question 4b: sunlight facing the Sun

The question

If the Earth receives 1368 W m^{-2} solar radiation at the equator, how much is reflected back into space, and how much is absorbed by the Earth?

Method. Multiply the incoming radiation by the fractions of Eq. (17.1):

$$\text{reflected} = a \times S = 0.30 \times 1368 = 410.4 \text{ W m}^{-2},$$
$$\text{absorbed} = (1 - a) \times S = 0.70 \times 1368 = 957.6 \text{ W m}^{-2}.$$

Check: $410.4 + 957.6 = 1368$ W m^{-2}. Every watt is accounted for.

Answer to Question 4b

Reflected: $0.3 \times 1368 = \mathbf{410.4}\,\mathrm{W\,m^{-2}}$. Absorbed: $0.7 \times 1368 = \mathbf{957.6}\,\mathrm{W\,m^{-2}}$.

A question that comes up

Is 1368 $\mathrm{W\,m^{-2}}$ really what arrives "at the equator"?
Not quite. The solar constant is the sunlight falling on a surface *at right angles to the Sun's rays, at the top of the atmosphere*, at the Earth's average distance from the Sun (Kopp and Lean, 2011). It arrives at the equator only at noon when the Sun is overhead there (around the equinoxes), and then only above the atmosphere: some is absorbed and scattered on the way down. Today's satellite value is 1361 $\mathrm{W\,m^{-2}}$; the 1368 is an older figure, and the difference changes no conclusion.

17.4 Question 4b′: why the average is only 341 $\mathrm{W\,m^{-2}}$

The question

The average incoming solar radiation for planet Earth is only 341 $\mathrm{W\,m^{-2}}$. Why do you think that is?

The 1368 $\mathrm{W\,m^{-2}}$ is for a surface squarely facing the Sun. Averaged over the whole Earth and over day and night, each square metre receives a quarter of that, for two reasons that each contribute a factor of two:

1. **Night.** At any moment half the Earth faces away from the Sun and receives nothing.
2. **Slanting rays.** On the day side, only the point directly under the Sun receives the full beam. Everywhere else the Sun is lower in the sky and its rays strike the ground at a slant, spreading the same energy over a larger area (this is Question 5, Chapter 18). Averaged over the sunlit hemisphere, this halves the energy per square metre.

Together: the Earth intercepts sunlight over a disc of area πR^2 but has a surface area of $4\pi R^2$ (Figure 16.1), so

$$\bar{S} = \frac{S}{4} = \frac{1368}{4} = 342\ \mathrm{W\,m^{-2}}. \tag{17.2}$$

where

$\bar{S}$	incoming solar radiation averaged over the whole Earth and over day and night ($\mathrm{W\,m^{-2}}$)	calculated
S	solar constant ($\mathrm{W\,m^{-2}}$)	1368, given

Answer to Question 4b′

The 1368 $\mathrm{W\,m^{-2}}$ falls only on a surface facing the Sun directly. At any time **half the Earth is in darkness**, and over the sunlit half **the Sun's rays arrive at a slant** except where the Sun is overhead, spreading the energy over a larger area. The Earth catches sunlight on a disc (πR^2) but spreads it over a sphere ($4\pi R^2$), so the global average is a quarter: $1368/4 = 342\,\mathrm{W\,m^{-2}}$, about $341\,\mathrm{W\,m^{-2}}$.

Why 341 and not 342? Global energy budgets built from satellite measurements give 341.3 $\mathrm{W\,m^{-2}}$ (Trenberth et al., 2009), a quarter of the solar constant they used (1365 $\mathrm{W\,m^{-2}}$). The difference from 342 is only in the value of the solar constant.

17.5 Question 4c: the global average

The question

If we take the average incoming solar radiation, then how much is reflected back into space, and how much is absorbed by the Earth?

Method. The same fractions, Eq. (17.1), applied to the average of $341\,\mathrm{W\,m^{-2}}$:

$$\text{reflected} = 0.30 \times 341 = 102.3\ \mathrm{W\,m^{-2}},$$
$$\text{absorbed} = 0.70 \times 341 = 238.7\ \mathrm{W\,m^{-2}}.$$

Answer to Question 4c

Reflected: $0.3 \times 341 = \mathbf{102.3}\,\mathrm{W\,m^{-2}}$. Absorbed: $0.7 \times 341 = \mathbf{238.7}\,\mathrm{W\,m^{-2}}$.

Check. Satellite-based budgets give about $102\,\mathrm{W\,m^{-2}}$ reflected and $239\,\mathrm{W\,m^{-2}}$ absorbed (Trenberth et al., 2009): these simple numbers reproduce the measured global budget.

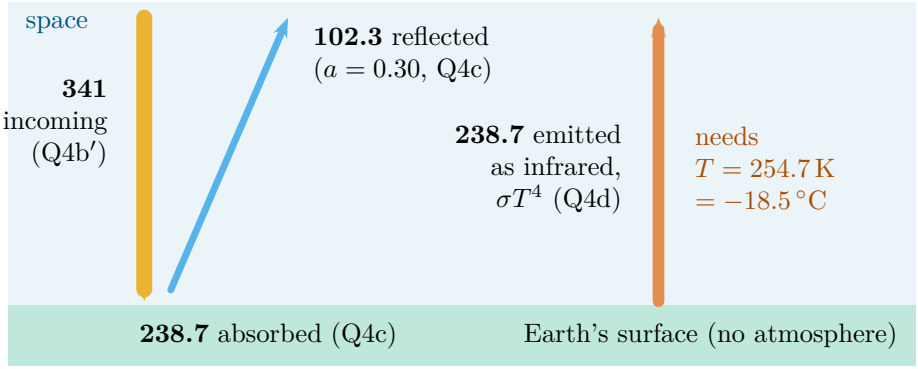


Figure 17.1. The energy flows of Question 4, in $\mathrm{W\,m^{-2}}$ averaged over the whole Earth, for an Earth without an atmosphere. Arrow widths are proportional to the flows. In balance, the infrared emitted equals the sunlight absorbed; the temperature needed to emit it is the answer to Question 4d.

17.6 Question 4d: the temperature of an airless Earth

The question

Earth must lose exactly as much energy as it gains, therefore the Earth must emit all the solar radiation it absorbs. However, if there is no atmosphere this happens very quickly (there is no greenhouse effect). First reorganise the Stefan–Boltzmann equation so that you can calculate the temperature in terms of the intensity of radiation ($T = \ldots$). Now calculate what the temperature of the Earth would be if all the radiation it absorbed from the Sun was released at the same time (use the data you calculated in 4c). Convert your answer from kelvin to Celsius.

Rearranging the law. Start from Eq. (14.1), $E = \sigma T^4$, and isolate T in two moves:

Step 1. Divide both sides by σ: $T^4 = E/\sigma$.

Step 2. Take the fourth root of both sides (the inverse of raising to the fourth power):

$$T = \left(\frac{E}{\sigma}\right)^{1/4}. \tag{17.3}$$

where

T	temperature of the emitting surface (K)	calculated
E	radiation emitted per square metre ($\mathrm{W\,m^{-2}}$)	238.7, from Question 4c
σ	Stefan–Boltzmann constant	$5.67 \times 10^{-8}\,\mathrm{W\,m^{-2}\ K^{-4}}$

Calculation. In balance the Earth emits exactly what it absorbs, so $E = 238.7\,\mathrm{W\,m^{-2}}$:

Step 3. $E/\sigma = 238.7/5.67 \times 10^{-8} = 4.2099 \times 10^{9}\,\mathrm{K^4}$.

Step 4. First square root: $\sqrt{4.2099 \times 10^{9}} = 64\,884$; second square root: $\sqrt{64\,884} = 254.7\,\mathrm{K}$.

Step 5. In Celsius: $254.7 - 273.16 = -18.5\,^{\circ}\mathrm{C}$.

Answer to Question 4d

$T = (E/\sigma)^{1/4}$. $T = (238.7/5.67 \times 10^{-8})^{1/4} = \mathbf{254.7\,K}$, which is $\mathbf{-18.5}\,^{\circ}\mathrm{C}$.

Check. This is the same calculation as the Earth's row of Question 3c, written differently: there $T = [1368 \times 0.7/(4\sigma)]^{1/4} = 254.9\,\mathrm{K}$. The 0.2 K difference comes only from using 341 instead of $1368/4 = 342\,\mathrm{W\,m^{-2}}$. And it is the temperature of curve A in Question 3 (254.1 K): curve A is the spectrum of an Earth without a greenhouse effect.

17.7 Question 4e: what the atmosphere does

The question

The actual average global temperature of the Earth between 1961 and 1990 is 14.0 °C (World Meteorological Organization). What difference does the atmosphere make to the global temperature of the Earth?

The size of the effect. The difference between the actual and the airless temperature is

$$14.0 - (-18.5) = 32.5\ ^{\circ}\mathrm{C},$$

which is also a difference of 32.5 K, since a step of one degree is the same size on both scales. The atmosphere keeps the Earth's surface about **33** °C **warmer** than it would otherwise be: above the freezing point of water instead of well below it.

How it does it. The atmosphere treats the two kinds of radiation differently (Figure 17.2):

1. **Sunlight (shortwave) passes through** largely unabsorbed and warms the surface.
2. **The surface emits infrared (longwave)**, Chapter 13, peaking near 10 μm.
3. **Greenhouse gases absorb much of that infrared**: water vapour, carbon dioxide, methane, nitrous oxide and ozone. These are molecules made of three or more atoms (or of two different atoms) whose vibrations and rotations can absorb and emit infrared; the main gases of the air, nitrogen and oxygen, cannot.
4. **The warmed atmosphere radiates in all directions**, including back down to the surface. The surface therefore receives the sunlight *plus* this downward infrared, about $333\,\mathrm{W\,m^{-2}}$ on average (Trenberth et al., 2009), and settles at a higher temperature.
5. **The planet as a whole still balances.** What escapes to space comes mostly from the cold upper atmosphere, and it still totals about $239\,\mathrm{W\,m^{-2}}$: the Earth seen from space radiates like a body at 255 K, even though its surface is at 288 K.

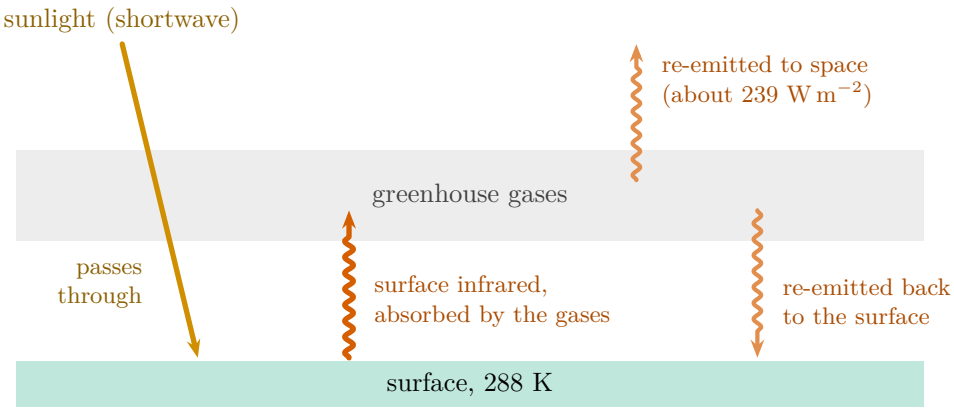


Figure 17.2. The greenhouse effect. Sunlight passes through the atmosphere and warms the surface; the surface's infrared is absorbed by greenhouse gases, which radiate both to space and back down. The extra downward infrared keeps the surface warmer than the 255 K the planet radiates at from space.

Answer to Question 4e

Without an atmosphere the Earth would average −18.5 °C; with it, 14.0 °C. The atmosphere's **greenhouse effect** raises the average surface temperature by about **32.5** °C (about 33 K). Greenhouse gases absorb the infrared emitted by the surface and radiate part of it back down, so the surface receives more energy than sunlight alone provides and stays at a habitable temperature, above the freezing point of water.

What greenhouse gases trap

Greenhouse gases act mainly on the *infrared* emitted by the Earth, not on incoming sunlight, which passes through largely unabsorbed. And they do not stop energy escaping: in balance the Earth still sends to space exactly as much as it absorbs (238.7 W m^{-2}). What they change is *where* it is emitted from: from the cold upper atmosphere instead of the warm surface. Say: "greenhouse gases absorb the infrared emitted by the surface and re-emit part of it back down, warming the surface by about 33 K".

Worked example: a one-layer greenhouse

Model the atmosphere as a single layer that lets all sunlight through and absorbs all the surface's infrared. In balance, the layer must emit 238.7 W m^{-2} to space; it emits the same amount downward, so the surface receives 238.7 + 238.7 = 477.4 W m^{-2} and must emit that:

$$T_{\text{surface}} = \left(\frac{2 \times 238.7}{5.67 \times 10^{-8}}\right)^{1/4} = 2^{1/4} \times 254.7 = 303 \text{ K } (30 \text{ °C}).$$

This overshoots the real 288 K, because the real atmosphere lets some infrared escape directly through the "atmospheric window" near 8–12 μm and because rising air and evaporation carry heat up from the surface. But it shows the mechanism, and why a thicker greenhouse atmosphere, like that of Venus (§ 16.7), makes the surface hotter still.

A question that comes up

Why is it called a "greenhouse" effect?
By analogy, and the analogy is loose. A real greenhouse stays warm mainly because its glass stops the warm air inside from rising away and mixing with cooler air. The atmosphere warms

the surface by radiation, as described above.

17.8 Question 4f: day and night without an atmosphere

The question

What effect would no atmosphere have on the variation in temperature on the Earth over the course of a day, compared with an Earth with an atmosphere?

Without an atmosphere, nothing would slow the loss of heat at night or soften the heating by day.

- **At night** the surface would radiate straight to space, with no downward infrared from greenhouse gases and cloud to replace part of the loss (§ 17.7), so it would cool very quickly and very far.
- **By day** there would be no cloud to reflect sunlight and no air to absorb part of it, and no winds to carry heat away, so the sunlit surface would heat strongly.
- **The atmosphere (and the ocean beneath it) also stores and moves heat**: from day to night and from warm regions to cold ones. Without it, each place would respond only to its own sunshine.

The Moon shows the result: at the same distance from the Sun but with no atmosphere, its equatorial surface swings from near 390 K at midday to below 100 K before dawn (Williams et al., 2017), while the Earth's global average daily range is only about 10 K (Williams, 2024).

Answer to Question 4f

The **daily range of temperature would be far larger**: very hot days and very cold nights. The atmosphere reduces the range by radiating infrared back to the surface at night (and through cloud), by reflecting and absorbing part of the sunlight by day, and by storing and moving heat. The airless Moon swings by about 300 K between day and night.

17.9 Question 4g: the assumptions

The question

What assumption have we made about the solid Earth for this calculation?

Every model rests on assumptions; naming them is part of using it well. For Question 4d we assumed that the Earth:

1. **has no source of heat of its own**: all its energy comes from the Sun. The heat flowing from the Earth's interior is real but tiny, under 0.1 $\mathrm{W\,m^{-2}}$ on average (Davies and Davies, 2010), so this assumption costs almost nothing (compare Jupiter, § 16.7);
2. **is a perfect black body**, emitting σT^4 with an emissivity of 1 (§ 12.6);
3. **has one uniform temperature** everywhere, by day and by night (in reality the equator and the poles differ by tens of degrees);
4. **is in equilibrium**, emitting at once all it absorbs and storing nothing (the question's "released at the same time");

5. **keeps an albedo of 0.30** even without an atmosphere. In fact much of that reflection is by cloud, which an airless Earth would not have; with a Moon-like albedo of about 0.1 the airless Earth would be nearer 270 K (Practice Set 16, item 16.3).

Answer to Question 4g

That the solid Earth has **no internal source of heat**, so that all the energy it radiates comes from the Sun; and that it behaves as a **black body at a single, uniform temperature in equilibrium**, radiating away immediately everything it absorbs.

The Earth does radiate

The assumption is that the Earth has no internal heat *source*: it radiates only energy that came from the Sun. It does not mean the Earth gives off no radiation: in this calculation it radiates 238.7 $\mathrm{W\,m^{-2}}$.

17.10 Try it yourself: Practice Set 17

Practice Set 17: the energy balance

Use $\sigma = 5.67 \times 10^{-8}\,\mathrm{W\,m^{-2}\ K^{-4}}$ and $°\mathrm{C} = \mathrm{K} - 273.16$.

17.1 Repeat Questions 4b′ to 4d with today's solar constant, 1361 $\mathrm{W\,m^{-2}}$, and an albedo of 0.30.

17.2 If the Earth's albedo rose from 0.30 to 0.32 (more cloud, or more ice), what would its black-body temperature become?

17.3 The Earth's surface at 14 °C emits 385.5 $\mathrm{W\,m^{-2}}$ (Question 2b), but only 238.7 $\mathrm{W\,m^{-2}}$ leaves the planet. How large is the difference, and what is it a measure of?

17.4 Using the one-layer greenhouse of §17.7, what would the surface temperature be? Why is the real surface cooler than this?

Answers to Practice Set 17

17.1 $\bar{S} = 1361/4 = 340.25\,\mathrm{W\,m^{-2}}$; absorbed $0.7 \times 340.25 = 238.2\,\mathrm{W\,m^{-2}}$; $T = (238.2/5.67 \times 10^{-8})^{1/4} = 254.6\,\mathrm{K} = -18.6\,°\mathrm{C}$, Eqs. (17.2) and (17.3).

17.2 $T = [1368 \times 0.68/(4 \times 5.67 \times 10^{-8})]^{1/4} = 253.1\,\mathrm{K}$, about 1.8 K colder. Small changes in albedo matter, which is why cloud and ice feedbacks are important in climate change.

17.3 $385.5 - 238.7 = 146.8\,\mathrm{W\,m^{-2}}$. This is the infrared emitted by the surface that does not escape to space: a measure of the greenhouse effect in energy terms.

17.4 $2^{1/4} \times 254.7 = 303\,\mathrm{K}$ (30 °C). The real atmosphere is not opaque at all infrared wavelengths (the atmospheric window lets some escape directly), and convection and evaporation carry heat upwards from the surface.

17.11 The energy balance in Python

These functions compute the equations of this chapter, and the book's tests check them against its answers.

```
def absorbed(incoming_wm2, albedo):
    """The part of the incoming radiation that is not reflected (W per square metre)."""
    return incoming_wm2 * (1 - albedo)
```

```
def temperature_from_emission(e_wm2):
    """Stefan-Boltzmann law turned round: temperature (K) from emission."""
    return (e_wm2 / SIGMA) ** 0.25
```

`absorbed(341, 0.3)` returns 238.7.

Chapter 18

Sun angle, solar intensity and the seasons

This chapter answers Questions 5a, 5b and 5c.

18.1 A spread beam is a weaker beam

Shine a torch straight down at a table: it makes a small, bright circle. Tilt it, and the same light spreads into a long, dim oval. The torch gives out the same energy; it is spread over more table, so each square centimetre receives less. Sunlight does the same. When the Sun is overhead its rays strike the ground squarely and their energy is concentrated; when it is low they strike at a slant and the same energy is spread over a larger area. The **solar intensity**, the energy received per unit area of ground, therefore depends on the angle of the Sun.

18.2 Measuring the spreading

Two angles describe the Sun's position (Figure 18.1):

- the **sun angle** β (also called the solar elevation or altitude), measured up from the horizon: 90° when the Sun is overhead, 0° when it is on the horizon;
- the **zenith angle** θ_z, measured from the vertical (the zenith, the point straight overhead): 0° when the Sun is overhead.

They add up to a right angle:

$$\beta = 90^\circ - \theta_z. \tag{18.1}$$

where

β	sun angle: height of the Sun above the horizon (degrees)	calculated
θ_z	zenith angle: angle of the Sun from the vertical (degrees)	Eq. (18.4)

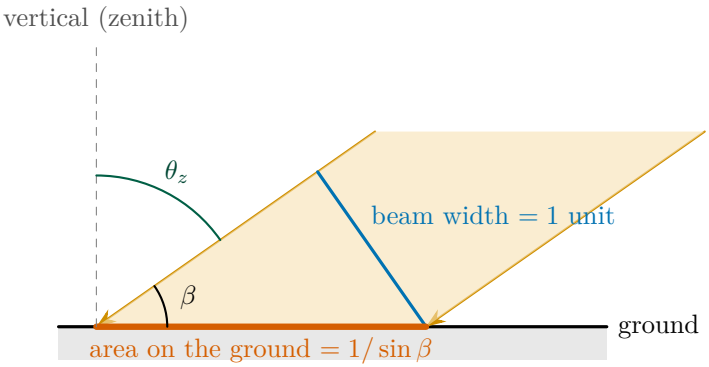


Figure 18.1. Beam spreading. A beam one unit wide, arriving at sun angle β, covers a strip of ground of length $1/\sin\beta$: the hypotenuse of a right-angled triangle whose side opposite the angle β is the beam width (Appendix B.2.4). Drawn here for $\beta = 35^\circ$, where the beam covers 1.74 units.

The length of ground covered by a beam one unit wide is called here the "surface area". From the right-angled triangle in Figure 18.1 (derived in Appendix B.2.4):

$$A = \frac{1}{\sin\beta}. \tag{18.2}$$

where

A	area over which a beam of unit width spreads ("surface area", "beam spreading"; units)	calculated
β	sun angle (degrees)	Eq. (18.1)

When the Sun is overhead, $\beta = 90°$, $\sin 90° = 1$ and $A = 1$ unit: the beam's full energy, 100 %, falls on one unit. In general the energy falling on each unit of ground is

$$I = \frac{100\,\%}{A} = 100\,\% \times \sin\beta. \tag{18.3}$$

where

I	solar intensity, as a percentage of the overhead value, per unit of ground (% per unit)	calculated
A	beam spreading from Eq. (18.2)	

Make sure your calculator is in **degree** mode: sin(83.5) should give 0.9936.

18.3 Where the noon Sun is overhead

Because the Earth's axis is tilted by 23.5° to its orbit, the latitude at which the noon Sun is directly overhead moves through the year (Figure 18.2). That latitude is the Sun's **declination** δ: +23.5° (the Tropic of Cancer) at the June solstice, 0° (the equator) at the equinoxes, and −23.5° (the Tropic of Capricorn) at the December solstice. At solar noon, the Sun's zenith angle at latitude φ is the difference between the two latitudes:

$$\theta_z = |\varphi - \delta|. \tag{18.4}$$

where

θ_z	noon zenith angle (degrees)	calculated
φ	latitude of the site (degrees, positive north)	30 or 60, given
δ	declination: latitude where the noon Sun is overhead (degrees)	+23.5 in June, −23.5 in December

For a site north of the Tropic of Cancer this gives two simple rules:

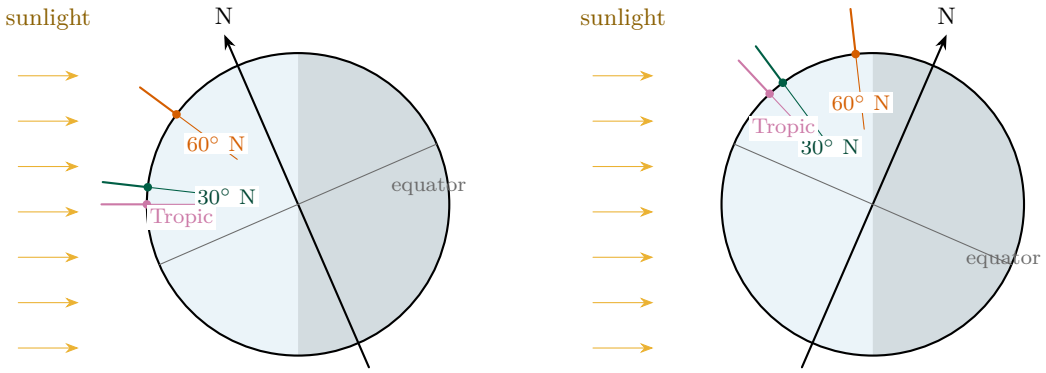
June 21 (summer solstice): $\theta_z = \varphi - 23.5°$; December 22 (winter solstice): $\theta_z = \varphi + 23.5°$.

The reason is geometric: walking north by one degree of latitude tilts your local vertical one degree further from the Sun's direction (Appendix B.2.5).

18.4 Question 5a: sketches of the noon Sun

The question

Using this information, draw a simple series of sketches for solar noon on June 21st at a site at 30° N, and on December 22nd and June 21st for a site at 60° N. Indicate the zenith angle,



21 June: $\delta = +23.5°$ **22 December:** $\delta = -23.5°$

Figure 18.2. The Earth at the two solstices, with sunlight arriving from the left. The short coloured lines are the local verticals at the Tropic of Cancer, 30° N and 60° N. In June (left) the northern hemisphere is tilted towards the Sun, the Sun is overhead at the Tropic, and the verticals at 30° N and 60° N are only 6.5° and 36.5° from the Sun's direction. In December (right) they are 53.5° and 83.5° from it, Eq. (18.4).

sun angle and beam spreading in each diagram.

Method. For each site and date, apply Eqs. (18.4), (18.1), (18.2) and (18.3) in turn. The calculations are collected in Table 18.1; the 30° N December case is not asked for in 5a but is needed for 5c, so it is included.

Table 18.1. Question 5 for the four cases. Zenith angle from Eq. (18.4), sun angle from Eq. (18.1), beam spreading from Eq. (18.2) and intensity from Eq. (18.3).

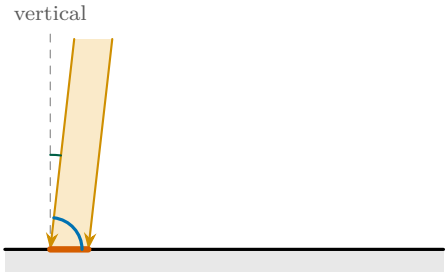
Site	Date	Zenith θ_z	Sun angle β	Spread $1/\sin\beta$	Intensity
30° N	21 June	$30 - 23.5 = 6.5°$	83.5°	$1/0.9936 = 1.006$	99.4 %
60° N	21 June	$60 - 23.5 = 36.5°$	53.5°	$1/0.8039 = 1.244$	80.4 %
60° N	22 Dec	$60 + 23.5 = 83.5°$	6.5°	$1/0.1132 = 8.834$	11.3 %
30° N	22 Dec	$30 + 23.5 = 53.5°$	36.5°	$1/0.5948 = 1.681$	59.5 %

Spreading in units of ground per unit of beam width; intensity in % of the overhead value per unit of ground.

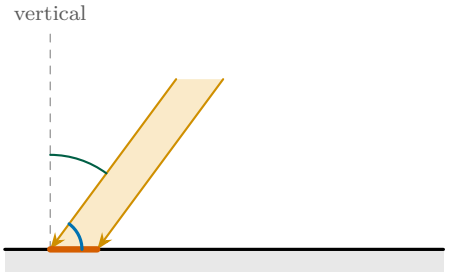
Answer to Question 5a

30° N, 21 June: zenith angle 6.5°, sun angle 83.5°, beam spreading $1/\sin 83.5° =$ **1.006 units**.
60° N, 21 June: zenith angle 36.5°, sun angle 53.5°, beam spreading $1/\sin 53.5° =$ **1.24 units**.
60° N, 22 December: zenith angle 83.5°, sun angle 6.5°, beam spreading $1/\sin 6.5° =$ **8.83 units**.
Sketches as in Figure 18.3.

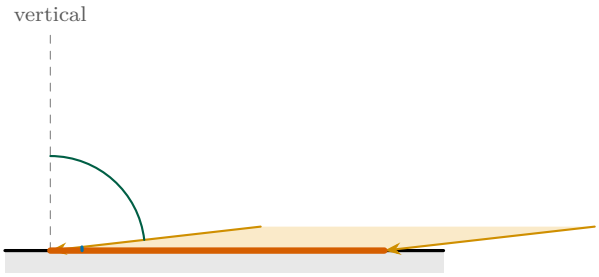
Check. The higher the Sun, the closer the spreading is to 1 unit; with the Sun just 6.5° above the horizon, a beam one unit wide is smeared across almost nine units of ground. In each case $\theta_z + \beta = 90°$.



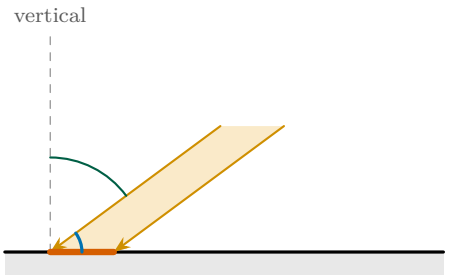
30° N, 21 June
zenith angle $\theta_z = 6.5°$ (green arc)
sun angle $\beta = 83.5°$ (blue arc)
spreading = 1.01 units (red strip)
intensity 99.4 % per unit



60° N, 21 June
zenith angle $\theta_z = 36.5°$ (green arc)
sun angle $\beta = 53.5°$ (blue arc)
spreading = 1.24 units (red strip)
intensity 80.4 % per unit



60° N, 22 December
zenith angle $\theta_z = 83.5°$ (green arc)
sun angle $\beta = 6.5°$ (blue arc)
spreading = 8.83 units (red strip)
intensity 11.3 % per unit



30° N, 22 December
zenith angle $\theta_z = 53.5°$ (green arc)
sun angle $\beta = 36.5°$ (blue arc)
spreading = 1.68 units (red strip)
intensity 59.5 % per unit (needed for 5c)

Figure 18.3. Question 5a: the noon Sun at each site, drawn to the correct angles and to one scale. Each beam (orange) is one unit wide and the red strip is the ground it covers. The sun angle β (blue arc) is measured from the ground, and the zenith angle θ_z (green arc) from the vertical.

18.5 Question 5b: why the sun angle makes the seasons

The question

Given these sketches, explain why the sun angle causes seasonal temperature changes in the mid-latitudes.

As the Earth orbits the Sun with its axis tilted, the latitude where the noon Sun is overhead moves between the tropics (Figure 18.2). At a mid-latitude site the noon Sun is therefore high in summer and low in winter: at 60° N, 53.5° in June and only 6.5° in December. A high Sun concentrates its beam on a small area and a low Sun spreads it over a large one, so each square metre of ground receives far more energy in summer (80 % per unit) than in winter (11 % per unit). More energy absorbed means warmer ground and air.

Answer to Question 5b

Because the Earth's axis is tilted, the noon **sun angle** at a mid-latitude site **rises in spring and summer and falls in autumn and winter**. A high sun angle concentrates the beam on a small area (little beam spreading, high intensity); a low sun angle spreads the same energy over a large area (much spreading, low intensity). So the ground receives more energy per square metre in summer and warms, and less in winter and cools. At 60° N the noon

intensity in June is about seven times that in December.

A question that comes up

Is the sun angle the only reason summers are warmer?
No; it is the main one, and two others work with it. **Day length:** at 60° N the Sun is up for about 18 hours at the June solstice and about 6 in December, so summer days add up much more energy. **Path through the atmosphere:** a low Sun's rays cross far more air, so more is scattered and absorbed before reaching the ground. All three effects point the same way. The distance from the Sun is *not* a reason: the Earth is closest to the Sun in early January, in the northern winter (Lutgens et al., 2019, ch. 2).

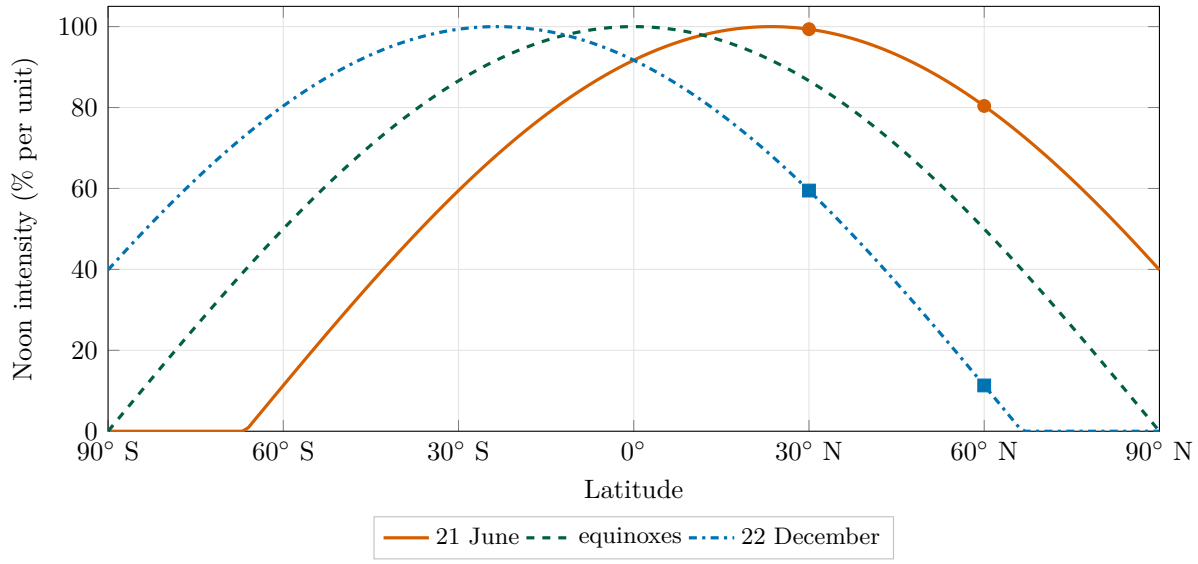


Figure 18.4. Noon solar intensity, Eq. (18.3), against latitude at the solstices and equinoxes. The dots are the four cases of Table 18.1. The peak follows the overhead Sun between the tropics; north of the Arctic Circle (66.5° N) the December curve reaches zero: the noon Sun does not rise.

18.6 Question 5c: comparing solar intensities

The question

Given your answers: (i) what is the percentage difference in solar intensity between a site at 30° N and 60° N during the winter solstice? (Hint: calculate the solar intensity as % per unit for each site first.) (ii) What is the percentage difference in solar intensity between a site at 60° N for the winter solstice compared to the summer solstice?

Step 1: the intensities, from Eq. (18.3) and Table 18.1:

$$I_{30\text{N, Dec}} = 100/1.681 = 59.5 \text{ \% per unit},$$
$$I_{60\text{N, Dec}} = 100/8.834 = 11.3 \text{ \% per unit},$$
$$I_{60\text{N, Jun}} = 100/1.244 = 80.4 \text{ \% per unit}.$$

Step 2: compare them. "Percentage difference" between two percentages can mean two things, and it is worth giving both:

- the difference in **percentage points**: subtract one percentage from the other;
- the **relative** difference: divide one by the other.

(i) 30° N against 60° N in December:

$59.5 - 11.3 = 48.2$ percentage points; $\dfrac{59.5}{11.3} = 5.25$ times; $\dfrac{59.5 - 11.3}{59.5} = 81\,\%$ less at 60°N.

(ii) 60° N, December against June:

$80.4 - 11.3 = 69.1$ percentage points; $\dfrac{80.4}{11.3} = 7.10$ times; $\dfrac{80.4 - 11.3}{11.3} = 610\,\%$ more in June.

Answer to Question 5c

(i) At the winter solstice 30° N receives **59.5 %** per unit and 60° N **11.3 %** per unit: a difference of **48.2 percentage points**. 30° N receives about **5.3 times** as much as 60° N (60° N receives 81 % less).

(ii) At 60° N the winter solstice gives **11.3 %** per unit and the summer solstice **80.4 %** per unit: a difference of **69.1 percentage points**. The summer noon intensity is about **7.1 times** the winter one.

Check. Both comparisons agree with Figure 18.4: the December curve falls steeply between 30° N and 60° N, and at 60° N the June and December curves are far apart.

Percentage points

A difference between two percentages is in *percentage points*. The summer and winter intensities at 60° N differ by 69.1 points; in relative terms the summer intensity is 7.1 times the winter one, which is 610 % more, not 69 %.

18.7 Try it yourself: Practice Set 18

Practice Set 18: sun angle

18.1 Coleraine is at 55.1° N. Find the noon zenith angle, sun angle, beam spreading and intensity at the June and December solstices.

18.2 What is the noon intensity at the equator at either solstice? And at the equinoxes?

18.3 What is the noon intensity at 60° N at the equinoxes, when the Sun is overhead at the equator?

18.4 What happens at the Arctic Circle (66.5° N) at noon on 22 December?

18.5 At which latitude is the noon intensity 100 % on 21 June, and why?

Answers to Practice Set 18

18.1 June: $\theta_z = 55.1 - 23.5 = 31.6°$, $\beta = 58.4°$, $A = 1/\sin 58.4° = 1.174$ units, $I = 85.2\,\%$ per unit. December: $\theta_z = 55.1 + 23.5 = 78.6°$, $\beta = 11.4°$, $A = 5.06$ units, $I = 19.8\,\%$ per unit. The June noon Sun gives Coleraine 4.3 times the December intensity.

18.2 At either solstice $\theta_z = 23.5°$, $\beta = 66.5°$, $A = 1.090$, $I = 91.7\,\%$ per unit. At the equinoxes the Sun is overhead: 100 %. The equator's noon intensity hardly changes through the year, which is why the tropics have little temperature seasonality.

18.3 $\theta_z = 60°$, $\beta = 30°$, $A = 1/\sin 30° = 2$ units, $I = 50\,\%$ per unit.

18.4 $\theta_z = 66.5 + 23.5 = 90°$, so $\beta = 0°$: the noon Sun is on the horizon, the beam spreading is infinite and the intensity is zero. North of this, the Sun does not rise at all on that day: the polar night.

18.5 At the Tropic of Cancer, 23.5° N, where $\theta_z = 0°$ and the Sun is directly overhead: the declination on 21 June is +23.5°, Eq. (18.4).

18.8 Sun angle in Python

These functions compute the equations of this chapter, and the book's tests check them against its answers.

```
def zenith_angle(latitude_deg, declination_deg):
    """Noon zenith angle (degrees) at a latitude for a solar declination."""
    return abs(latitude_deg - declination_deg)
```

```
def sun_angle(zenith_deg):
    """Sun angle above the horizon (degrees)."""
    return 90.0 - zenith_deg
```

```
def beam_spreading(sun_deg):
    """Length of ground covered by a beam one unit wide."""
    return 1.0 / math.sin(math.radians(sun_deg))
```

```
def intensity_percent(sun_deg):
    """Solar intensity as a percentage per unit of ground."""
    return 100.0 / beam_spreading(sun_deg)
```

`intensity_percent(sun_angle(zenith_angle(60, -23.5)))` returns about 11.3.

Chapter 19

Putting Part II together

19.1 One chain of ideas

The five questions are one argument, each step resting on the one before:

1. Everything above absolute zero radiates, over a spread of wavelengths described by the Planck curve (Chapter 12).
2. The **peak** of that spread moves to shorter wavelengths as temperature rises (Wien's law): the Sun, at about 5800 K, peaks in visible light; the Earth, near 288 K, in the thermal infrared (Question 1, Chapter 13).
3. The **total** emission rises as the fourth power of temperature (Stefan–Boltzmann law): twenty times the temperature means 166 000 times the emission (Question 2, Chapter 14).
4. A planet settles at the temperature at which it emits as much as it absorbs. That temperature depends on its distance from the Sun (inverse-square law) and its albedo, and can be read from its spectrum (Question 3, Chapters 15 and 16).
5. Where the actual temperature is higher than this, an atmosphere is trapping infrared (the greenhouse effect: Venus, the Earth) or the planet has its own heat (Jupiter). For the Earth the greenhouse effect is worth about 33 K (Question 4, Chapter 17).
6. Across the Earth's surface, the energy received depends on the angle of the Sun. The tilt of the axis moves that angle through the year and so makes the seasons (Question 5, Chapter 18).

This heating explains the structure met in Part I: a troposphere heated from below and a stratosphere heated by ozone (Part I, § 5.4), thickest over the tropics (Part I, § 7.4). Part III shows how the uneven heating from equator to pole fills the air with moisture and sets it in motion.

19.2 All the answers in one table

Table 19.1. Every answer of Part II, with the section where it is worked out.

Q	Answer	Where
1	$T_{\text{Sun}} = 2898/0.5 = 5796\,\text{K} = 5522.8\,^{\circ}\text{C}$	§ 13.3
2a	$E_{\text{Sun}} = 6.40 \times 10^{7}\,\text{W m}^{-2}$	§ 14.3
2b	$E_{\text{Earth}} = 385.5\,\text{W m}^{-2}$ (at 287.16 K)	§ 14.4
2c	$5796/287.16 = 20.2$, about 20 times hotter	§ 14.5
2d	$6.40 \times 10^{7}/385.5 = 166\,000$ times more emission	§ 14.6
2e	$E \propto T^{4}$: $20.18^{4} = 166\,000$ (a power law)	§ 14.7
3a	peaks at 12, 14 and 16 μm (A, B, C)	§ 15.3
3b	$T_A = 241.5\,\text{K}$, $T_B = 207.0\,\text{K}$, $T_C = 181.1\,\text{K}$ (true curves: 254.1, 210.0, 184.1 K)	§ 15.4

Q	Answer	Where
3c	Mercury 439, Venus 185, Earth 255, Mars 210, Jupiter 110 K	§ 16.5
3d	A Earth, B Mars, C Venus	§ 16.6
3e	(i) Mars: further away, 43 % of the Earth's sunlight; (ii) Venus: albedo 0.9; (iii) Venus 737 K, intense CO_2 greenhouse; (iv) Mercury: no atmosphere (Mars nearly none)	§ 16.7
4a	70 % absorbed	§ 17.2
4b	reflected 410.4, absorbed 957.6 W m^{-2}	§ 17.3
4b′	night and slanting rays: disc πR^2 against sphere $4\pi R^2$, so $1368/4 = 342 \approx 341$ W m^{-2}	§ 17.4
4c	reflected 102.3, absorbed 238.7 W m^{-2}	§ 17.5
4d	$T = (E/\sigma)^{1/4} = 254.7$ K $= -18.5$ °C	§ 17.6
4e	greenhouse effect warms the surface by about 32.5 K (33 K)	§ 17.7
4f	much larger day–night range: very hot days, very cold nights	§ 17.8
4g	no internal heat source; black body at one temperature, in equilibrium	§ 17.9
5a	30° N June: $\theta_z = 6.5°$, $\beta = 83.5°$, 1.006 units; 60° N June: 36.5°, 53.5°, 1.24 units; 60° N December: 83.5°, 6.5°, 8.83 units	§ 18.4
5b	tilt: high summer Sun concentrates the beam, low winter Sun spreads it	§ 18.5
5c	(i) 59.5 against 11.3 % per unit: 48.2 points, 5.3 times; (ii) 80.4 against 11.3 % per unit: 69.1 points, 7.1 times	§ 18.6

19.3 Mistakes to avoid

- **Using Celsius in a radiation law.** σT^4 and C/T need kelvin. Convert first, and convert back only at the end (§ 14.5).
- **Forgetting the fourth root.** When solving $E = \sigma T^4$ for T, divide by σ *and then* take the fourth root, Eq. (17.3); a single square root gives a value about 65 000 instead of 255.
- **Mixing up the sun angle and the zenith angle.** The sun angle is from the horizon, the zenith angle from the vertical; they add to 90°, Eq. (18.1). Beam spreading uses the sun angle.
- **Calculator in radians.** sin(83.5) in radian mode gives 0.97 instead of 0.9936.
- **Subtracting percentages and calling the result a percentage.** Two intensities of 80.4 % and 11.3 % differ by 69.1 percentage points; in relative terms one is 7.1 times the other (§ 18.6).
- **Reading the peak of a coarsely sampled spectrum as exact.** The highest data point lies near, not at, the peak (§ 15.4).

19.4 Self-test

Self-test: one line each

S1. Why is the Sun's radiation called shortwave and the Earth's longwave?

S2. If a star is three times as hot as the Sun, how does its total emission per square metre compare?

S3. Why does Venus have a lower black-body temperature than the Earth but a far higher surface temperature?

S4. Why do we divide the solar constant by four?

S5. Without an atmosphere, would the Earth be colder by day, by night, or both?

S6. Why is there little seasonal change in temperature at the equator?

Answers to the self-test

S1. Wien's law: the Sun is about 20 times hotter, so its emission peaks at a wavelength about 20 times shorter (0.5 against 10 μm) (Chapter 13).

S2. $3^4 = 81$ times as much, Eq. (14.2).

S3. Its bright clouds reflect most sunlight (high albedo), so it absorbs little; but its thick CO_2 atmosphere traps the infrared from the surface (§ 16.7).

S4. The Earth catches sunlight on a disc of area πR^2 but spreads it over a sphere of area $4\pi R^2$ (Figure 16.1).

S5. Mostly by night: nights would be far colder. Days could be hotter, as on the Moon (§ 17.8). On average it would be about 33 K colder.

S6. The noon Sun is always high there (at least 66.5°), so the intensity changes little through the year (Practice Set 18, item 18.2).

Appendix B

Part II: formulas, derivations, data and symbols

B.1 Formulas

Every formula used in this part, with the equation number where it is introduced and explained. Temperatures are in kelvin; angles in degrees.

Quantity	Formula	Where, and what for
Speed of a wave	$c = f\lambda$	Eq. (12.1); frequency from wavelength
Energy of a photon	$\varepsilon = hc/\lambda$	Eq. (12.2); why short waves carry more energy
Planck curve	$B_\lambda = \frac{2hc^2}{\lambda^5}\frac{1}{e^{hc/\lambda k_B T} - 1}$	Eq. (12.3); the shape of every spectrum
Wien's law	$\lambda_{\max} = C/T$	Eq. (13.1); peak from temperature (Q1, Q3)
Wien's law, rearranged	$T = C/\lambda_{\max}$	Eq. (13.2); temperature from peak (Q1, Q3b)
Stefan–Boltzmann law	$E = \sigma T^4$	Eq. (14.1); total emission (Q2)
Ratio of emissions	$E_1/E_2 = (T_1/T_2)^4$	Eq. (14.2); Q2e
Stefan–Boltzmann, rearranged	$T = (E/\sigma)^{1/4}$	Eq. (17.3); temperature from emission (Q4d)
Inverse-square law	$S = S_E/d^2$	Eq. (16.1); sunlight at each planet (Q3c)
Planetary temperature	$T = \left[\frac{S(1-a)}{4\sigma}\right]^{1/4}$	Eq. (16.2); black-body temperature (Q3c)
Absorbed fraction	$1 - a$	Eq. (17.1); Q4a–c
Global average sunlight	$\bar{S} = S/4$	Eq. (17.2); Q4b′
Sun angle	$\beta = 90° - \theta_z$	Eq. (18.1); Q5a
Noon zenith angle	$\theta_z = \lvert\varphi - \delta\rvert$, $\delta = \pm 23.5°$ at the solstices	Eq. (18.4); Q5a
Beam spreading	$A = 1/\sin\beta$	Eq. (18.2); Q5a
Solar intensity	$I = 100\,\%/A = 100\,\% \times \sin\beta$	Eq. (18.3); Q5c
Kelvin to Celsius	$T_{\circ\mathrm{C}} = T_{\mathrm{K}} - 273.16$	§ 11.2

B.2 Derivations for the curious

These derivations show where the formulas of this part come from. None is needed to answer the questions; each is referred to from the chapter where its result is used.

B.2.1 Wien's law from the Planck curve

The peak of the Planck curve, Eq. (12.3), is where its slope is zero. Write $x = hc/(\lambda k_B T)$, so that a short wavelength is a large x. Up to constants that do not depend on λ, Eq. (12.3) becomes

$$B_\lambda \propto \frac{x^5}{e^x - 1}.$$

Setting the derivative with respect to x to zero:

$$\frac{d}{dx}\left(\frac{x^5}{e^x-1}\right) = \frac{5x^4(e^x-1) - x^5 e^x}{(e^x-1)^2} = 0 \quad \Longrightarrow \quad 5(e^x-1) = xe^x \quad \Longrightarrow \quad x = 5\,(1-e^{-x}).$$

This equation has no algebraic solution, but it is easy to solve by repeated substitution: start with $x = 5$, then $x = 5(1-e^{-5}) = 4.966$, then $5(1-e^{-4.966}) = 4.965$, and it settles at $x = 4.965$. Turning back to the wavelength,

$$\lambda_{\max} = \frac{hc}{4.965\,k_B\,T} = \frac{C}{T}, \quad \text{with}$$
$$C = \frac{(6.626\times10^{-34})(2.998\times10^{8})}{4.965\times1.381\times10^{-23}} = 2.8978\times10^{-3}\ \mathrm{m\,K} = 2897.8\ \mu\mathrm{m\,K},$$

which is Eq. (13.1) with $C = 2898\,\mu\mathrm{m\,K}$. (Since x falls steadily as λ rises, the peak found in terms of x is the same point as the peak in terms of λ; and λ^{-5} in Eq. (12.3) is a constant times x^5, which is where the x^5 comes from.)

B.2.2 The Stefan–Boltzmann law from the Planck curve

The total power leaving one square metre of a black body is π times the spectral radiance, Eq. (12.3), added up over all wavelengths (the factor π collects the radiation leaving in every direction above the surface):

$$E = \pi \int_0^\infty B_\lambda(T)\,d\lambda.$$

With the same substitution $x = hc/(\lambda k_B T)$ the integral becomes

$$E = \frac{2\pi k_B^4 T^4}{h^3c^2}\int_0^\infty \frac{x^3}{e^x-1}\,dx = \frac{2\pi k_B^4 T^4}{h^3c^2}\times\frac{\pi^4}{15}.$$

All the temperature dependence has come out as T^4, and the rest is the Stefan–Boltzmann constant:

$$\sigma = \frac{2\pi^5 k_B^4}{15\,h^3c^2} = 5.670\times10^{-8}\ \mathrm{W\,m^{-2}\,K^{-4}},$$

which is Eq. (14.1). The value of the integral, $\pi^4/15 = 6.494$, is a standard result.

B.2.3 The black-body temperature of a planet

This is worked through in full in § 16.4: absorbed sunlight on the disc, $S(1-a)\pi R^2$, equals emission from the sphere, $\sigma T^4 4\pi R^2$, which gives Eq. (16.2). Dividing Eq. (16.2) for two planets shows how temperature scales:

$$\frac{T_1}{T_2} = \left(\frac{S_1(1-a_1)}{S_2(1-a_2)}\right)^{1/4},$$

so with equal albedos, a planet at a quarter of the sunlight (twice the distance) is $4^{1/4} = \sqrt{2} = 1.41$ times colder.

B.2.4 Beam spreading

In Figure 18.1, draw a line across the beam at right angles to the rays: its length is the beam width, 1 unit. This line, the strip of ground covered by the beam (length A), and the edge ray make a right-angled triangle. The ground strip is the hypotenuse, and the beam width is the side opposite the angle β at which the rays meet the ground. By the definition of the sine,

$$\sin\beta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{1}{A} \qquad \Longrightarrow \qquad A = \frac{1}{\sin\beta},$$

which is Eq. (18.2). The intensity per unit of ground is the beam's energy divided by A, which is Eq. (18.3). In words: the energy per unit area is proportional to the sine of the sun angle, or equivalently to the cosine of the zenith angle, since $\sin\beta = \cos\theta_z$.

B.2.5 The noon zenith angle

The local vertical at latitude φ points along the Earth's radius, at an angle φ from the equatorial plane (Figure 18.2). At noon the Sun lies in the same north–south plane, in the direction at angle δ from the equatorial plane, where δ is the latitude at which it is overhead. The Sun's rays are parallel across the Earth, because the Sun is so far away, so the angle between the vertical and the direction of the Sun is the difference of the two angles:

$$\theta_z = |\varphi - \delta|,$$

which is Eq. (18.4). With $\delta = +23.5°$ (June) and $-23.5°$ (December), and a site north of the Tropic of Cancer, this gives $\theta_z = \varphi - 23.5°$ and $\theta_z = \varphi + 23.5°$. The value 23.5° is the tilt of the Earth's axis to the plane of its orbit (more precisely 23.44°).

B.3 The spectra of Question 3, recomputed

Each column of the Question 3 data (Table 15.1) was compared with the Planck curve, Eq. (12.3), by finding the temperature and the scale factor that minimise the relative differences between the two. All three columns fit one scale factor, about 316 times the spectral radiance in $\mathrm{W\,m^{-2}\,sr^{-1}\,\mu m^{-1}}$, and the best-fitting temperatures are

$$T_A = 254.1\ \mathrm{K}, \qquad T_B = 210.0\ \mathrm{K}, \qquad T_C = 184.1\ \mathrm{K},$$

the black-body temperatures of the Earth, Mars and Venus calculated in Question 3c (Table 16.1). Table B.1 sets the given values beside the fitted curves: they agree to within about 0.3 % everywhere, which confirms that the data are Planck curves sampled every 2 μm.

Because the true peaks fall between the 2 μm sampling points, the highest data point in each column lies 0.3–0.6 μm to the right of the true peak. That is why Question 3b, which reads the peak from the data, gives temperatures a few kelvin below the true ones (§ 15.4).

B.4 Symbols and constants

Table B.1. The data of Question 3 (“given”) beside the Planck curves for 254.1, 210.0 and 184.1 K on the same scale (“Planck”). Wavelengths in μm.

	Data A		**Data B**		**Data C**	
λ	given	Planck, 254.1 K	given	Planck, 210.0 K	given	Planck, 184.1 K
4	26.199	26.16	1.3338	1.338	0.11992	0.1202
6	386.28	385.9	53.056	53.18	10.648	10.67
8	970.65	969.8	218.86	219.2	65.619	65.7
10	1313.3	1312	398.14	398.6	151.8	151.9
12	1363.5	1363	502.49	503	224.7	224.9
14	1248.6	1248	527.86	528.3	264.23	264.4
16	1074.3	1074	502.41	502.8	273.39	273.5
18	896.25	895.8	452.58	452.9	262.51	262.6
20	737.1	736.7	395.22	395.4	241.03	241.1
22	603.1	602.8	339.25	339.4	215.38	215.5
24	493.45	493.2	288.62	288.8	189.36	189.4
26	404.91	404.7	244.61	244.7	164.92	165
28	333.76	333.6	207.16	207.2	142.9	142.9
30	276.58	276.5	175.67	175.7	123.56	123.6
True peaks, Eq. (13.1) with $C = 2897.8$:		A 11.40 μm,		B 13.80 μm,		C 15.74 μm.

Table B.2. Every symbol used in this part. Each equation’s own *where* list repeats the symbols it uses.

Symbol	**Meaning**	**Unit**	**First used**
a	albedo: fraction of sunlight reflected	–	Eq. (16.2)
A	beam spreading (the “surface area”)	units	Eq. (18.2)
B_λ	spectral radiance	$\mathrm{W\,m^{-2}\,sr^{-1}\,m^{-1}}$	Eq. (12.3)
β	sun angle (solar elevation)	degrees	Eq. (18.1)
c	speed of light	$\mathrm{m\,s^{-1}}$	Eq. (12.1)
C	Wien’s constant	$\mu\mathrm{m\,K}$	Eq. (13.1)
d	distance from the Sun	AU	Eq. (16.1)
δ	solar declination	degrees	Eq. (18.4)
E	emission per square metre (“intensity of radiation”)	$\mathrm{W\,m^{-2}}$	Eq. (14.1)
ϵ	emissivity	–	§ 12.6
ε	energy of one photon	J	Eq. (12.2)
f	frequency	Hz	Eq. (12.1)
h	Planck’s constant	J s	Eq. (12.2)
I	solar intensity per unit of ground	% per unit	Eq. (18.3)
k_B	Boltzmann’s constant	$\mathrm{J\,K^{-1}}$	Eq. (12.3)
λ	wavelength	m or μm	Eq. (12.1)
$\lambda_{\max}$	wavelength of peak emission	μm	Eq. (13.1)
φ	latitude	degrees	Eq. (18.4)
R	radius of a planet	m	§ 16.4
S	solar constant at a planet	$\mathrm{W\,m^{-2}}$	Eq. (16.1)
$\bar{S}$	sunlight averaged over the globe, day and night	$\mathrm{W\,m^{-2}}$	Eq. (17.2)

Symbol	Meaning	Unit	First used
σ	Stefan–Boltzmann constant	$\mathrm{W\,m^{-2}\ K^{-4}}$	Eq. (14.1)
T	absolute temperature	K	Eq. (13.1)
θ_z	zenith angle	degrees	Eq. (18.1)

Table B.3. Constants. Answers in this part use the rounded values; the precise values are given for reference.

Quantity	Value used	Precise value	Source
Wien's constant C	$2898\,\mu\mathrm{m\,K}$	$2897.77\,\mu\mathrm{m\,K}$	Tiesinga et al. (2021)
Stefan–Boltzmann σ	$5.67 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$	5.670374×10^{-8}	Tiesinga et al. (2021)
Planck's constant h	–	$6.62607015 \times 10^{-34}\,\mathrm{J\,s}$ (exact)	Tiesinga et al. (2021)
Speed of light c	–	$2.99792458 \times 10^{8}\,\mathrm{m\,s^{-1}}$ (exact)	Tiesinga et al. (2021)
Boltzmann's constant k_B	–	$1.380649 \times 10^{-23}\,\mathrm{J\,K^{-1}}$ (exact)	Tiesinga et al. (2021)
Solar constant S	$1368\,\mathrm{W\,m^{-2}}$	$1361\,\mathrm{W\,m^{-2}}$	Kopp and Lean (2011); Prša et al. (2016)
Earth's albedo	0.30	about 0.29–0.30	Forster et al. (2021); Williams (2024)
Kelvin–Celsius offset	273.16	273.15 (exact)	Tiesinga et al. (2021)
Earth's axial tilt	23.5°	23.44°	Stull (2017)
Sun's effective temperature	–	5772 K	Prša et al. (2016)

B.5 About the figures and the data

B.5.1 How the figures were made

Every figure in this part was drawn for it as vector graphics, in the document itself with TikZ and pgfplots, so it stays sharp at any size and prints clearly in black and white (each curve has its own line style or marker as well as its own colour).

- The **Planck curves** (Figures 12.3, 13.1 and 15.1) are computed from Eq. (12.3) with the constants of Table B.3.
- The **Question 3 graph** (Figure 15.1) plots the values of Table 15.1; the curves through them are the fits described in Appendix B.3.
- The **planetary temperatures** (Figure 16.2) are the values of Table 16.1: calculated with Eq. (16.2), and the actual averages.
- The **sun-angle sketches** (Figure 18.3) are drawn to the exact angles of Table 18.1 and to one scale, each beam one unit (5 mm) wide, so the lengths of the red strips can be compared directly.
- The **energy flows** (Figure 17.1) are the values of Questions 4b′–4d, with arrow widths proportional to the flows.

B.5.2 Limits to bear in mind

- The black-body model treats each planet as having one temperature, one albedo and no internal heat. It explains the main differences between the planets, not the details of any one of them.
- The sun-angle calculation gives the intensity at *noon* at the top of the atmosphere. Daily totals also depend on day length, and the energy reaching the ground on cloud and on the length of the path through the air (§ 18.5).
- Values for the planets (albedos, average temperatures) are rounded here; current measured values are in Williams (2024).

Appendix D

Constants, units and conversions

D.1 Constants

Each part works with the rounded values below. The precise values are given for reference.

Table D.1. Constants used in the book.

Quantity	Value used	Precise value	Source	Part
Acceleration due to gravity g	$9.81\,\mathrm{m\,s^{-2}}$	$9.80665\,\mathrm{m\,s^{-2}}$ (standard)	NOAA et al. (1976)	I, III
Gas constant for dry air R_d	$287\,\mathrm{J\,kg^{-1}\,K^{-1}}$	$287.05\,\mathrm{J\,kg^{-1}\,K^{-1}}$	Wallace and Hobbs (2006)	I, III
Specific heat of dry air c_p	$1004\,\mathrm{J\,kg^{-1}\,K^{-1}}$	$1004\,\mathrm{J\,kg^{-1}\,K^{-1}}$	Wallace and Hobbs (2006)	I
Standard sea-level pressure	1000 hPa (Part I)	1013.25 hPa	NOAA et al. (1976)	I
Standard sea-level temperature	15 °C	288.15 K	NOAA et al. (1976)	I
Stefan–Boltzmann constant σ	$5.67 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$	5.670374×10^{-8}	Tiesinga et al. (2021)	II
Wien's constant C	$2898\,\mu\mathrm{m\,K}$	$2897.77\,\mu\mathrm{m\,K}$	Tiesinga et al. (2021)	II
Planck's constant h	–	$6.62607015 \times 10^{-34}\,\mathrm{J\,s}$ (exact)	Tiesinga et al. (2021)	II
Speed of light c	–	$2.99792458 \times 10^{8}\,\mathrm{m\,s^{-1}}$ (exact)	Tiesinga et al. (2021)	II
Boltzmann's constant k_B	–	$1.380649 \times 10^{-23}\,\mathrm{J\,K^{-1}}$ (exact)	Tiesinga et al. (2021)	II
Solar constant S	$1368\,\mathrm{W\,m^{-2}}$	$1361\,\mathrm{W\,m^{-2}}$	Kopp and Lean (2011)	II
Earth's axial tilt	23.5°	23.44°	Stull (2017)	II
Earth's rotation rate Ω	$7.292 \times 10^{-5}\,\mathrm{s^{-1}}$	$7.292 \times 10^{-5}\,\mathrm{s^{-1}}$	Holton and Hakim (2013)	III
Kelvin–Celsius offset	273.15 (273.16 in Part II)	273.15 (exact)	Tiesinga et al. (2021)	all

D.2 Units and conversions

Quantity	**Conversion**
Pressure	1 hPa = 1 mb = 100 Pa; 1 mmHg (torr) = 1.333 hPa
Temperature	$T_{\mathrm{K}} = T_{\circ\mathrm{C}} + 273.15$; $T_{\circ\mathrm{C}} = \frac{5}{9}\,(T_{\circ\mathrm{F}} - 32)$
Temperature difference	1 K = 1 °C = 1.8 °F
Wind speed	$1\ \mathrm{kt} = 0.514\ \mathrm{m\,s^{-1}} = 1.852\ \mathrm{km\,h^{-1}}$
Length	1 km = 1000 m; $1\ \mu\mathrm{m} = 10^{-6}$ m
Power per area	$\mathrm{W\,m^{-2}}$: watts per square metre

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